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Previous Year Questions (GATE EE): Numerical Ability

(1 Mark Questions)

Year: GATE EE 2026

Question 1: At how many points will the curves $y = x^2$ and $y = -x^2 - 2x - 1$ intersect in the real $(x, .y)$ plane?

A: 0

B: 1

C: 2

D: 3

×

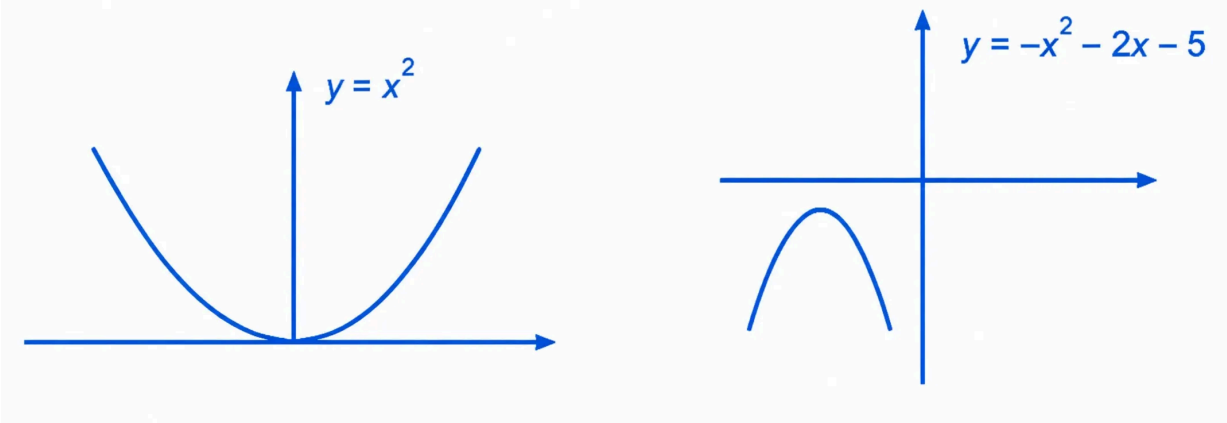
Are you liking this doc?

Yes

No

Answer: A

Solution:



No intersection points,

$y = x^2$ Always positive

$y = -x^2 - 2x - 5$ Always negative

Year: GATE EE 2025

Question 2: What is the value of $\left(\frac{3^{81}}{27^4}\right)^{1/3}$?

A: 3^{13}

B: 3^{96}

C: 3^{23}

D: 3^{69}

Answer: C

Solution:

$$\left(\frac{3^{81}}{27^4}\right)^{1/3} = \left(\frac{3^{81}}{(3^3)^4}\right)^{1/3} = \left(\frac{3^{81}}{3^{12}}\right)^{1/3}$$

$$= (3^{81-12})^{1/3} = (3^{69})^{1/3} = 3^{23}$$

Are you liking this doc?

Yes No

Year: GATE EE 2024

Question 3: The decimal number system uses the characters 0, 1, 2, ... , 8, 9, and the octal number system uses the characters 0, 1, 2, ... , 6, 7.

For example, the decimal number 12 (= 1 × 10^1 + 2 × 10^0) is expressed as 14 (= 1 ×

$8^1 + 4 \times 8^0$) in the octal number system.

The decimal number 108 in the octal number system is

A: 168

B: 108

C: 150

D: 154

Answer: D

Solution:

To convert the decimal number 108 to the octal number system, we follow a systematic division by 8, capturing the remainders which represent the octal digits.

Here's the step-by-step process:

1. Divide the decimal number by 8 and record the quotient and the remainder.

$108 \div 8 = 13\text{quotient}, 4\text{remainder}$

2. Next, take the quotient and divide it again by 8, recording the new quotient and the new remainder.

$13 \div 8 = 1\text{quotient}, 5\text{remainder}$

3. Repeat the process until the quotient is zero.

$1 \div 8 = 0\text{quotient}, 1\text{remainder}$

4. The octal number is formed by reading the remainders from the last obtained (from bottom to top).

So the remainders in order are $1, 5, 4$. This forms the octal number:

$\text{Octal Number} = 154$

Thus, the decimal number 108 in the octal number system is **154**.

Therefore, the correct answer is:

Option D **154**

Are you liking this doc?

👍 Yes 👎

Year: GATE EE 2024

Question 4: A shopkeeper buys shirts from a producer and sells them at 20% profit. A customer has to pay ₹3,186.00 including 18% taxes, per shirt. At what price did the shopkeeper buy each shirt from the producer?

A: ₹2,500.00

B: ₹1,975.40

C: ₹2,250.00

D: ₹2,548.80

Are you liking this doc?

Yes

No

Answer: C

Solution:

Let's break down the problem step by step to find the price the shopkeeper paid for each shirt.

1. **Calculate the price of the shirt before tax:**

The customer paid ₹3,186.00, which includes an 18% tax. Let 'x' represent the price of the shirt before tax. We can set up an equation:

x + 0.18x = ₹3,186.00

Combining like terms:

1.18x = ₹3,186.00

Solving for x:

x = ₹3,186.00 / 1.18

x = ₹2,700.00

Therefore, the price of the shirt before tax is ₹2,700.00

2. **Calculate the shopkeeper's selling price:**

The customer paid ₹2,700.00 for the shirt, which includes the shopkeeper's 20% profit. Let 'y' represent the shopkeeper's selling price.

We can set up another equation:

y + 0.20y = ₹2,700.00

Combining like terms:

1.20y = ₹2,700.00

Solving for y:

y = ₹2,700.00 / 1.20

y = ₹2,250.00

Therefore, the shopkeeper's selling price (before tax) was ₹2,250.00

Answer:

The shopkeeper bought each shirt from the producer for **₹2,250.00**. So the correct option is **C**.

Are you liking this doc?

Yes No

Year: GATE EE 2024

Question 5: If, for non-zero real variables x, y, and real parameter $a > 1$, $x : y = (a+1) : (a-1)$, then, the ratio $(x^2 - y^2) : (x^2 + y^2)$ is

A: $2a:(a^2+1)$

B: $a:(a^2+1)$

C: $2a:(a^2-1)$

D: $a:(a^2-1)$

Are you liking this doc?

Yes

No

Answer: A

Solution:

We are given the ratio between two variables x and y as follows:

$x: y = (a+1): (a-1)$

We can express this as:

$\frac{x}{y} = \frac{a+1}{a-1}$

From this, we can write:

$x = \frac{a+1}{a-1} \cdot y$

We need to find the ratio $\frac{x^2 - y^2}{x^2 + y^2}$.

First, let's express x^2 and y^2 in terms of y. Using $x = \frac{a+1}{a-1} \cdot y$, we get:

$x^2 = (\frac{a+1}{a-1})^2 y^2$

We already have y^2. Now, let's substitute these into the desired ratio:

$\frac{x^2 - y^2}{x^2 + y^2}$

Substitute x^2 and y^2:

$\frac{(\frac{a+1}{a-1})^2 y^2 - y^2}{(\frac{a+1}{a-1})^2 y^2 + y^2}$

Factor out y^2 from the numerator and the denominator:

$\frac{y^2 ((\frac{a+1}{a-1})^2 - 1)}{y^2 ((\frac{a+1}{a-1})^2 + 1)}$

The y^2 terms cancel out:

$\frac{(\frac{a+1}{a-1})^2 - 1}{(\frac{a+1}{a-1})^2 + 1}$

Now, let's simplify further. Consider $(\frac{a+1}{a-1})^2$:

$(\frac{a+1}{a-1})^2 = \frac{(a+1)^2}{(a-1)^2} = \frac{a^2 + 2a + 1}{a^2 - 2a + 1}$

So, the ratio becomes:

Are you liking this doc?

Yes No

$$\frac{\frac{a^2+2a+1}{a^2-2a+1}-1}{\frac{a^2+2a+1}{a^2-2a+1}+1}$$

Rewriting the numerator and the denominator by getting a common denominator:

$$\frac{\frac{((a^2+2a+1)-(a^2-2a+1))}{a^2-2a+1}}{\frac{((a^2+2a+1)+(a^2-2a+1))}{a^2-2a+1}}$$

This simplifies to:

$$\frac{\frac{4a}{a^2-2a+1}}{\frac{2a^2+2}{a^2-2a+1}}$$

Further simplifying:

$$\frac{4a}{2a^2+2}$$

Finally, we get:

$$\frac{4a}{2(a^2+1)}=\frac{2a}{a^2+1}$$

Thus, the ratio $(x^2-y^2):(x^2+y^2)$ is:

Option A: $2a:(a^2+1)$

Year: GATE EE 2023

Question 6: Given a fair six-faced dice where the faces are labelled '1', '2', '3', '4', '5', and '6', what is the probability of getting a '1' on the first roll of the dice and a '4' on the second roll?

A: $\frac{1}{36}$

B: $\frac{1}{6}$

C: $\frac{5}{6}$

D: $\frac{1}{3}$

Are you liking this doc?

Yes No

Answer: A

Solution:

Since the dice is fair, all the six faces have an equal probability of appearing on each roll. The probability of getting a '1' on the first roll is 1/6, and the probability of getting a '4' on the second roll is also $\frac{1}{6}$.

Since the two rolls are independent of each other, meaning that the outcome of one roll does not affect the outcome of the other roll, we can find the probability of both events happening together by multiplying their individual probabilities :

Probability of getting a '1' on the first roll and a '4' on the second roll

= Probability of getting a '1' on the first roll $\times$ Probability of getting a '4' on the second roll

= $\frac{1}{6} \times \frac{1}{6}$

= $\frac{1}{36}$

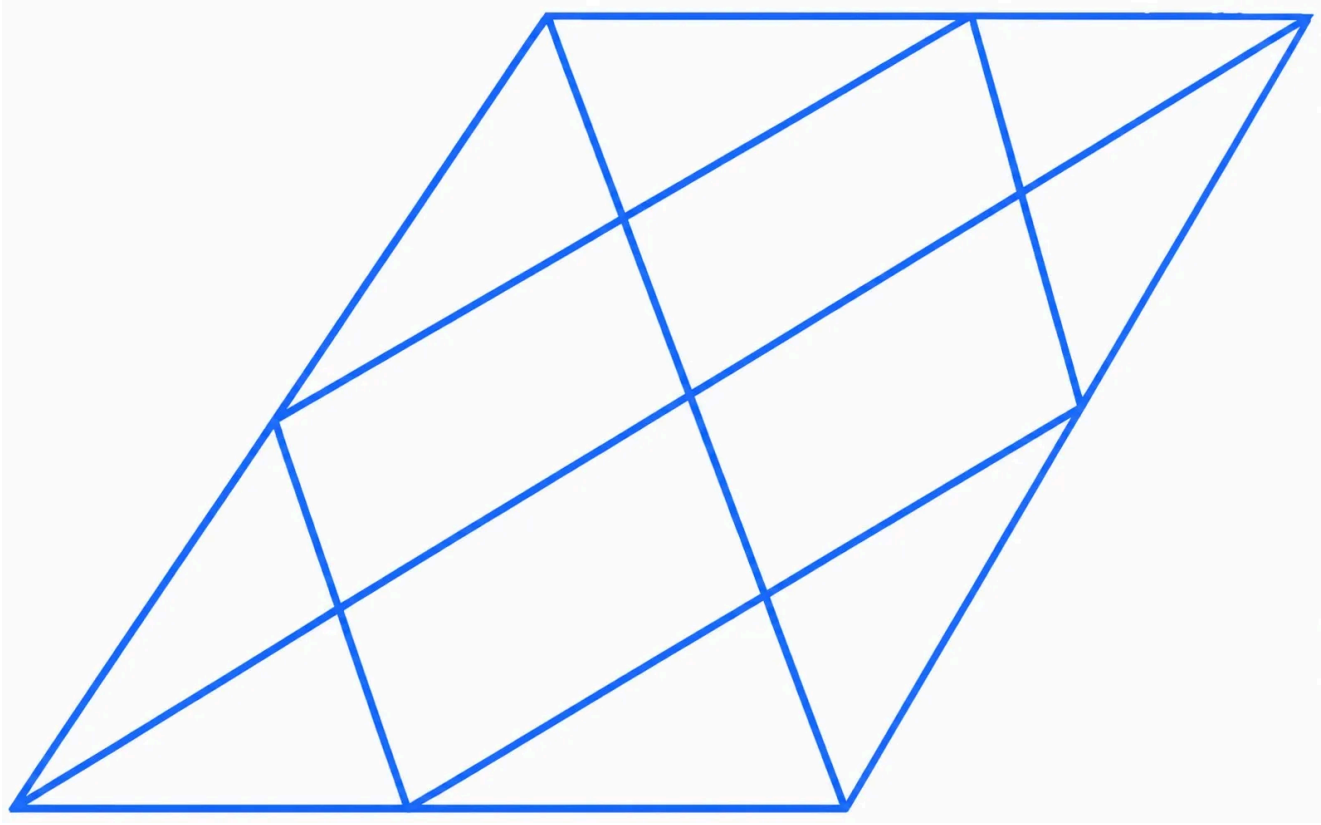
Therefore, the probability of getting a '1' on the first roll of the dice and a '4' on the second roll is $\frac{1}{36}$.

Year: GATE EE 2023

Question 7: How many triangles are present in the given figure?

Are you liking this doc?

Yes No



A: 12

B: 16

C: 20

D: 24

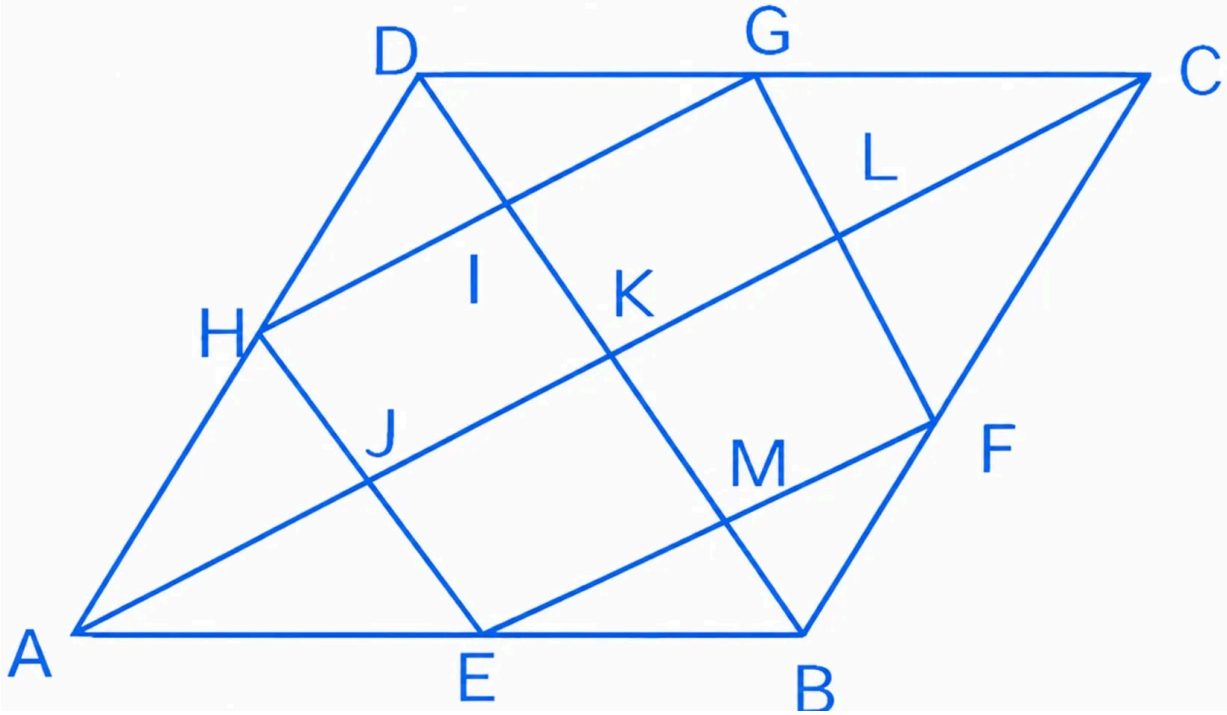
Are you liking this doc?

Yes

No

Answer: C

Solution:



Number of triangles $= \triangle ABC, \triangle ACD, \triangle BDC,$
$\triangle ABD, \triangle CDK, \triangle BCK, \triangle ADK, \triangle ABK, \triangle AHJ,$
$\triangle AEH, \triangle AEJ, \triangle BEM, \triangle BFM, \triangle CFL, \triangle CGL,$
$\triangle CGF, \Delta DGl, \triangle DHI, \Delta DGH.$

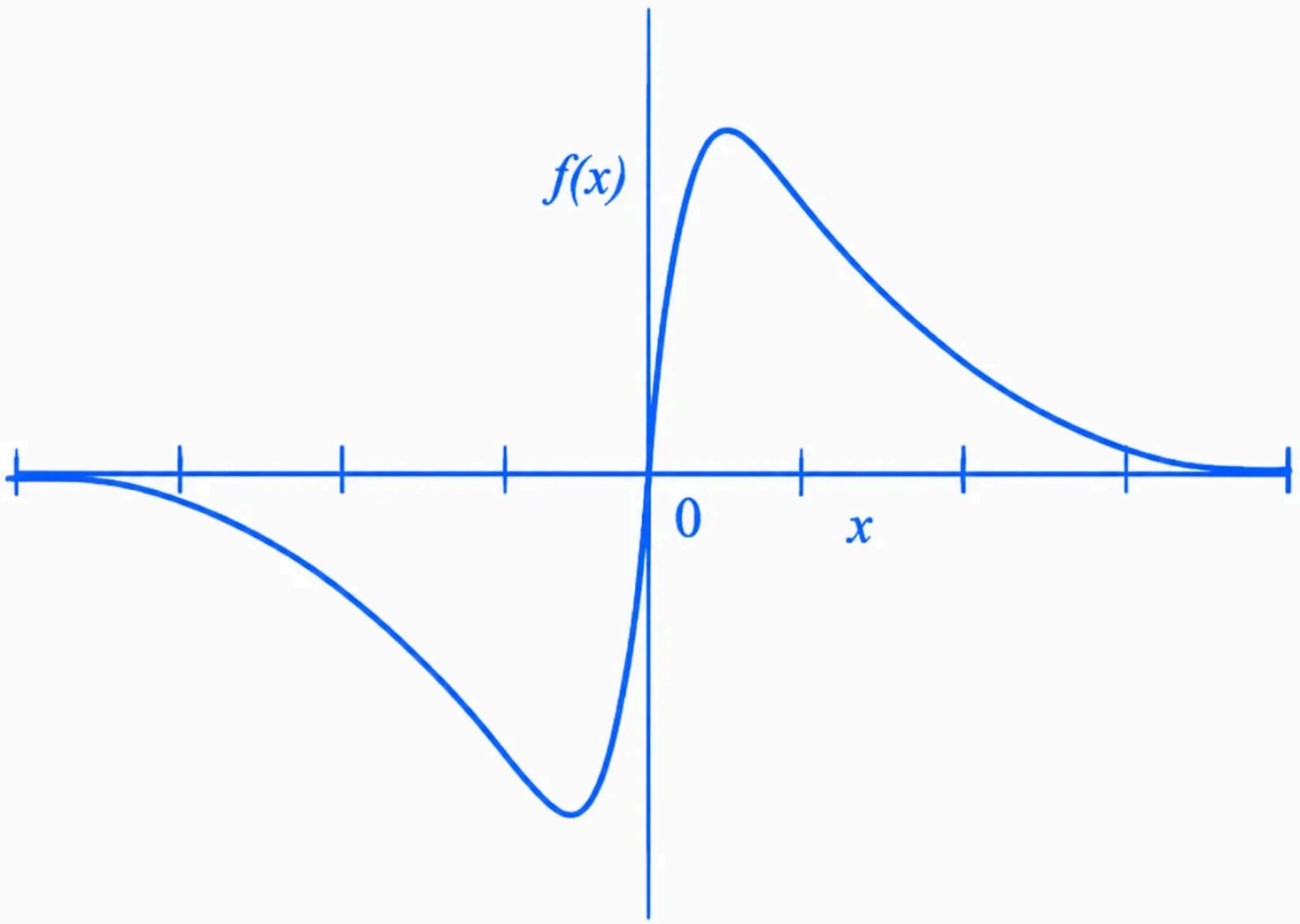
$\therefore$ Total number of triangles $= 20$

Year: GATE EE 2023

Question 8: Which of the following options represents the given graph?

Are you liking this doc?

Yes No



A: $f(x) = x^2\, 2^{-|x|}$

B: $f(x) = x2^{-|x|}$

C: $f(x) = |x|2^{-x}$

D: $f(x) = x2^{-x}$

Answer: B

Solution: The given function is odd function.

Property of odd function.

f(-x) = -f(x)

Therefore, option (B) is an odd function.

Year: GATE EE 2023

Question 9: the digit in the unit's place of the product $3^{999} \times 7^{1000}$ is _____.

A: 7

B: 1

C: 3

D: 9

Are you liking this doe?

Yes No

Answer: A

Solution:

To find the unit digit of the given product, we only need to consider the unit digit of each factor, because the other digits will not affect the unit digit of the product.

Let's analyze the unit digits of the two factors separately :

The units digit of 3^n has a repeating cycle of 4: 3, 9, 7, 1, 3, 9, 7, 1, ...

The units digit of 7^n has a repeating cycle of 4: 7, 9, 3, 1, 7, 9, 3, 1, ...

For powers of 3, the unit's digit follows the cycle: 3, 9, 7, 1. So, the unit's digit of 3^{999} will be the same as that of $3^{999 mod 4}$, which is $3^3 = 27$, and hence the unit's digit is 7.

For powers of 7, the unit's digit follows the cycle: 7, 9, 3, 1. So, the unit's digit of 7^{1000} will be the same as that of $7^{1000 mod 4}$, which is $7^0 = 1$, and hence the unit's digit is 1.

Multiplying the unit's digits of 3^{999} and 7^{1000} gives us $7 \times 1 = 7$.

Therefore, the digit in the unit's place of the product $3^{999} \times 7^{1000}$ is 7.

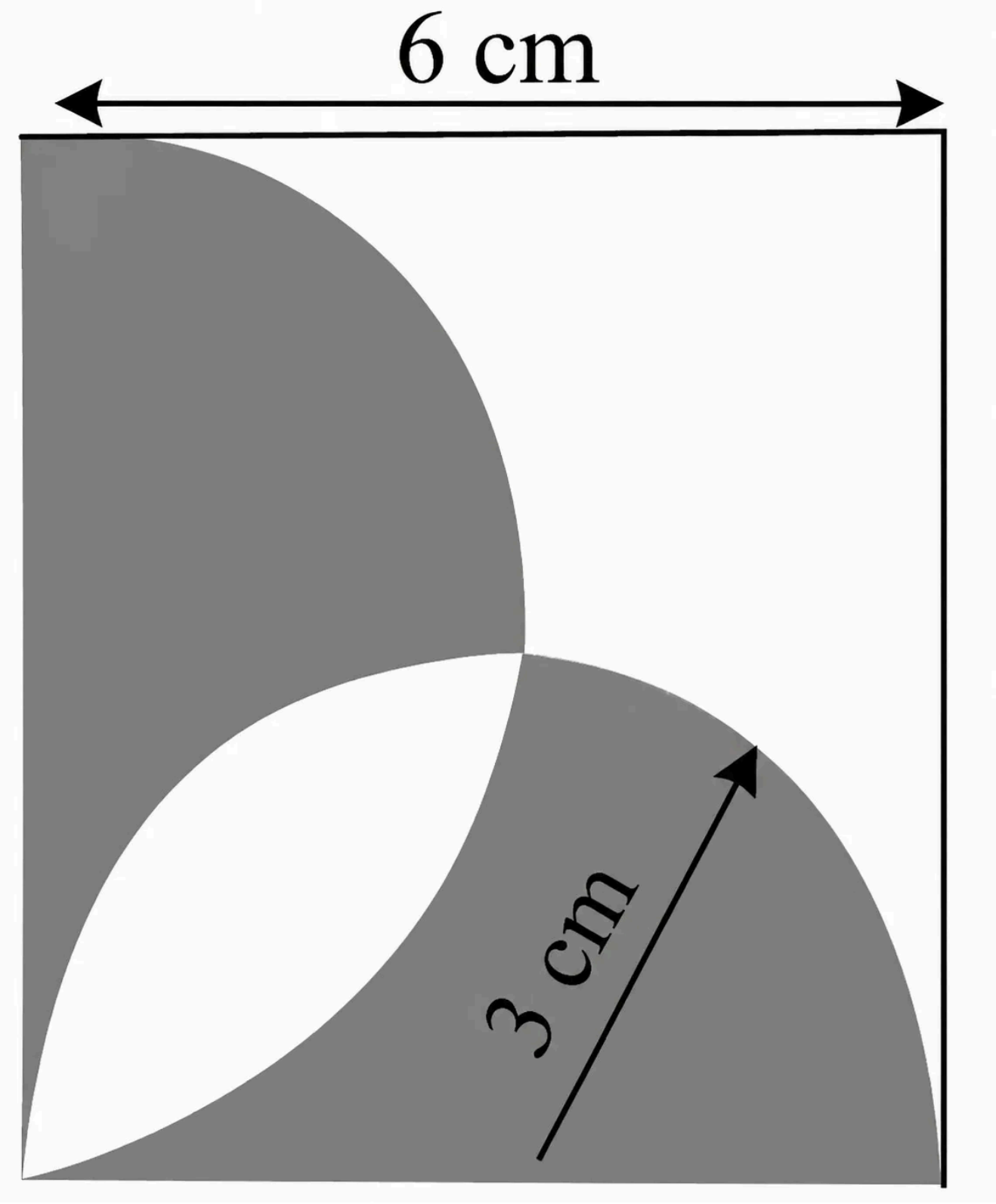
Year: GATE EE 2023

Question 10: A square with sides of length 6 cm is given. The boundary of the shaded region is defined by two semi-circles whose diameters are the sides of the square, as shown.

The area of the shaded region is _____ cm^2.

Are you liking this doc?

Yes No



A: 6π

B: 18

C: 20

D: 9π

Are you liking this doc?

Yes No

Answer: B

Solution:

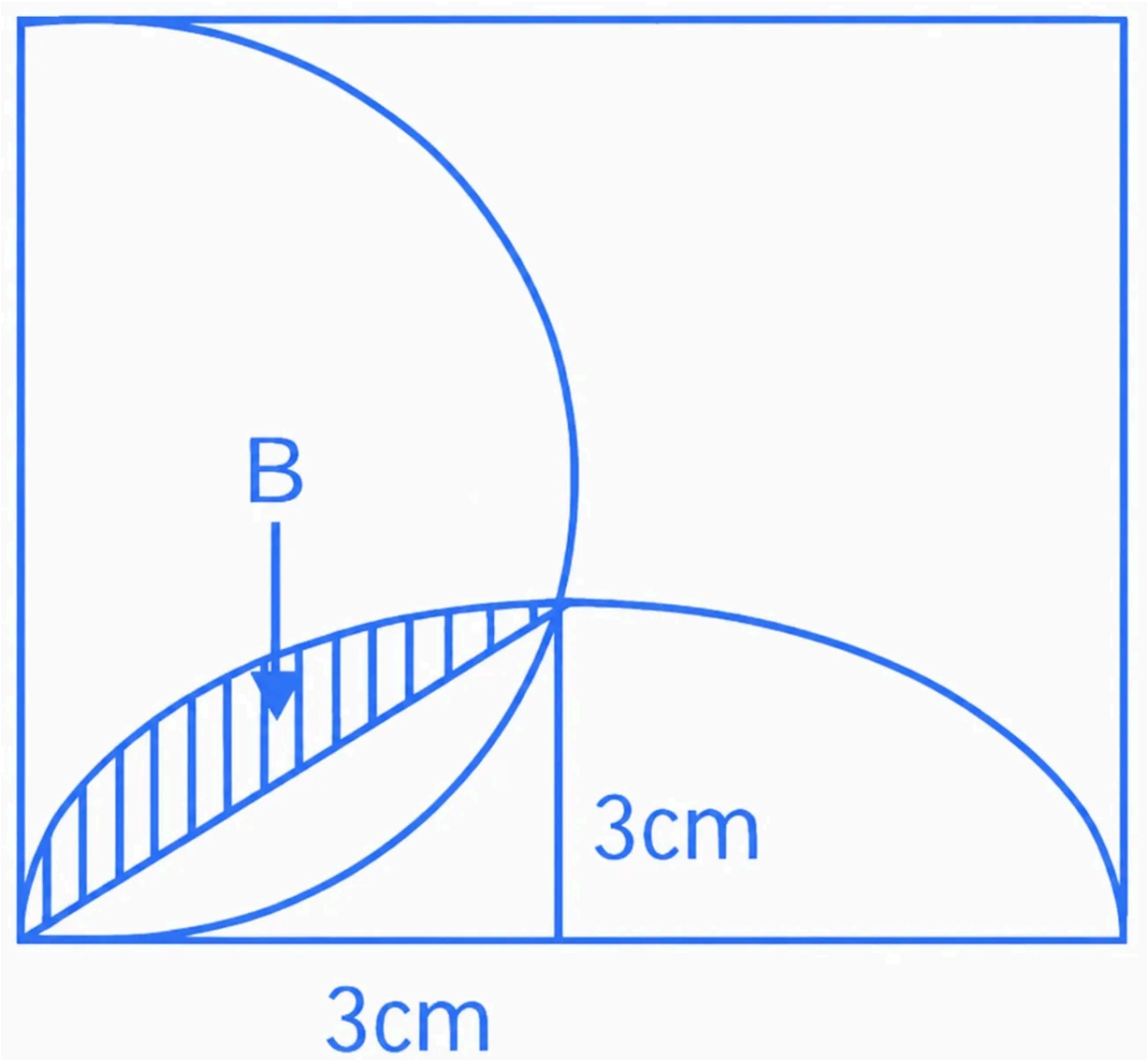
Area of shaded region

$$= \frac{\pi r^2}{2} + \frac{\pi r^2}{2} - 2A$$

$$= 8\pi - 2A[\because r = 3\ cm]$$

Are you liking this doc?

Yes No



$$\text{Area, } B = \frac{\pi r^2}{4} - \frac{1}{2} \times 3 \times 3$$

$$= \frac{9\pi}{4} - \frac{9}{2}$$

$$\therefore \text{Area of } A = 2B = \frac{9\pi}{2} - 9$$

Now, area of shaded region

$$= 9\pi - 2(\frac{9\pi}{2} - 9)$$

$$= 18\ cm^2$$

Are you liking this doc?

Yes No

Year: GATE EE 2022

Question 11: In a 500 m race, P and Q have speeds in the ratio of 2 : 1. Q starts the race when P has already covered 140 m. What is the distance between P and Q (in m) when P wins the race?

A: 20

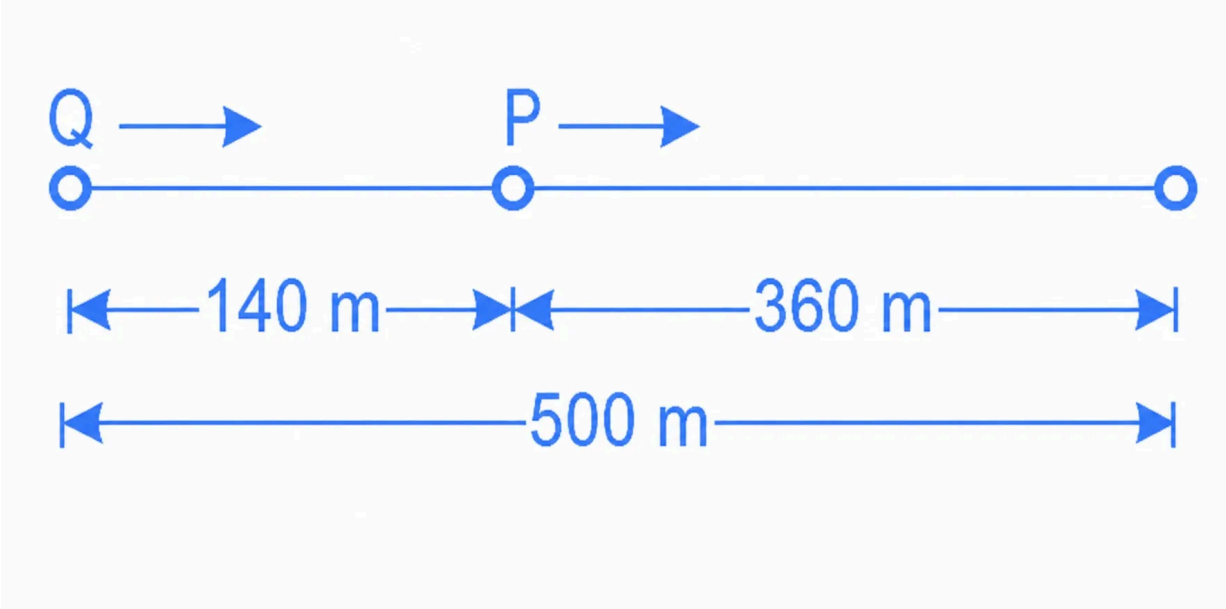
B: 40

C: 60

D: 140

Answer: A

Solution:



P : Q = 3 : 4

P = 3x

Q = 4x

So, 3x = 360

x = 120 m

Q = 4x = 4 × 120 = 480 m

When P reaches 500 metres and Q reaches only 480 metres.

∴ P wins by 20 m.

Year: GATE EE 2022

Question 12: Three bells P, Q and R are rung periodically in a school. P is rung every 20 minutes; Q is rung every 30 minutes and R is rung every 50 minutes. If all the three bells are rung at 12:00 P.M., when will the three bells ring together again the next time?

A: 5:00 P.M.

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Yes

No

B: 5:30 P.M.

C: 6:00 P.M.

D: 6:30 P.M.

Answer: A

Solution:

Three bells P, Q, R.

P = 20 minutes

Q = 30 minutes

R = 50 minutes

L.C.M. of 20, 30, 50 = 300 minutes = 5 hours

12:00 PM + 5 hours = 5:00 P.M.

So, three bells ring together again at 5:00 P.M.

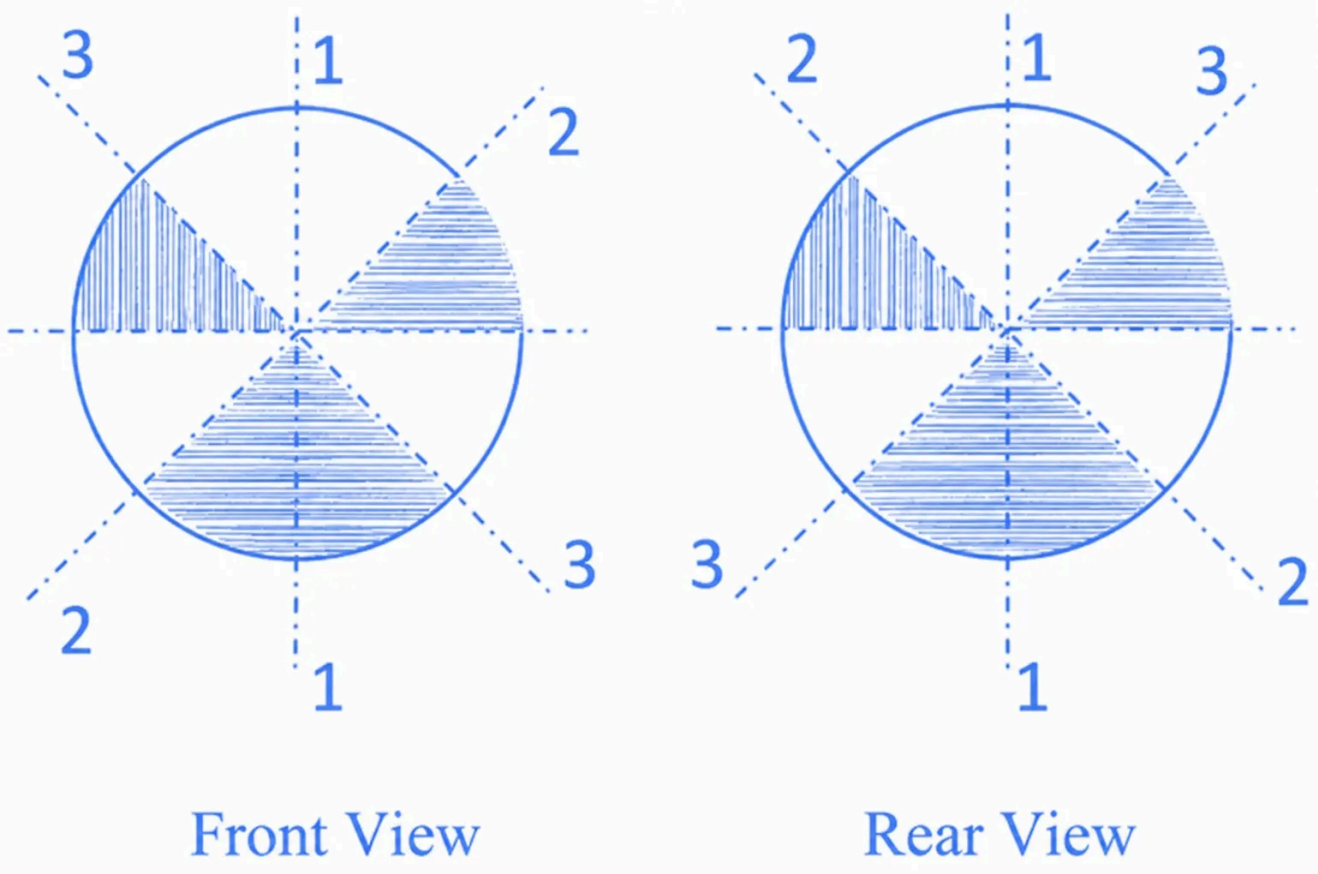
Year: GATE EE 2022

Question 13: The figure below shows the front and rear view of a disc, which is shaded with identical patterns. The disc is flipped once with respect to any one of the fixed axes 1-1, 2-2 or 3-3 chosen uniformly at random. What is the probability that the disc DOES NOT retain the same front and rear views after the flipping operation?

Are you liking this doc?

Yes

No



A: 0

B: $\frac{1}{3}$

C: $\frac{2}{3}$

D: 1

Answer: C

Solution:

n = Total = 3

m = Favourable = disc does not retain the same front and rear views after the flipping = 2

Probability = $\frac{m}{n} = \frac{2}{3}$

Year: GATE EE 2021

Question 14: For a regular polygon having 10 sides, the interior angle between the sides of the polygon, in degrees, is

A: 396

Are you liking this doc?

Yes No

B: 324

C: 216

D: 144

Answer: D

Solution:

Given : Sides of polygon, $n = 10$

Since, interior angle between the sides of the polygon

$$= \frac{(n-2)}{n} \times 180^\circ$$

$$= \frac{(10-2)}{10} \times 180^\circ = 144^\circ$$

Hence, the correct option is (D).

Year: GATE EE 2021

Question 15: Which one of the following numbers is exactly divisible by $(11^{13} + 1)$?

A: $11^{26} + 1$

B: $11^{33} + 1$

C: $11^{39} - 1$

D: $11^{52} - 1$

Are you liking this doc?

Yes No

Answer: D

Solution:

$\because \frac{(x^n - a^n)}{(x+a)}$ for n is even

$x = (11)^{13}$

$a = (1)^{13}$

$\therefore \frac{x^n - a^n}{(x+a)} = \frac{\{(11)^{13}\}^n - \{(1)^{13}\}^n}{(11)^{13} + (1)^{13}}$

$= \frac{\{(11)^{13}\}^4 - \{(1)^{13}\}^4}{(11)^{13} + (1)^{13}}$ for $n = 4$ (even)

$= \frac{[\{(11)^{13}\}^2 - \{(1)^{13}\}^2][\{(11)^{13}\}^2 + \{(1)^{13}\}^2]}{(11)^{13} + (1)^{13}}$

$\because a^2 - b^2 = (a-b)(a+b)$

$\frac{x^n - a^n}{(x+a)} = \frac{[(11^{13} - 1^{13})(11^{13} + 1^{13})][(11^{13})^2 + (1^{13})^2]}{11^{13} + 1^{13}}$

$\therefore (11^{52} - 1)$ is divisible by $(11^{13} + 1)$.

Hence, the correct option is (D).

Year: GATE EE 2019

Question 16: The missing number in the given sequence 343, 1331, _____, 4913 is

A: 3375

B: 2744

C: 2197

D: 4096

Answer: C

Are you liking this doc?

Yes No

Year: GATE EE 2019

Question 17: It takes two hours for a person X to mow the lawn. Y can mow the same lawn in four hours. How long (in minutes) will it take X and Y, if they work together to mow the lawn?

A: 120

B: 90

C: 60

D: 80

Answer: D

Year: GATE EE 2018

Question 18: Functions F(a, b) and G(a, b) are defined as follows:

F(a, b) = (a - b)^2 and G(a, b) = |a - b| , where |x| represents the absolute value of x.

What would be the value of G(F(1, 3), G(1, 3))?

A: 2

B: 4

C: 6

D: 36

Answer: A

Year: GATE EE 2018

Question 19: For what values of k given below is $\frac{(k+2)^2}{k-3}$ an integer?

A: 4, 8, 18

B: 4, 10, 16

C: 4, 8, 28

D: 8, 26, 28

Answer: C

Year: GATE EE 2018

Question 20: The three roots of the equation f(x) = 0 are x = {-2, 0, 3}. What are the three values of x for which f(x -3) = 0?

A: -5, -3, 0

Are you liking this doc?

Yes No

B: -2, 0, 3

C: 0, 6, 8

D: 1, 3, 6

Answer: D

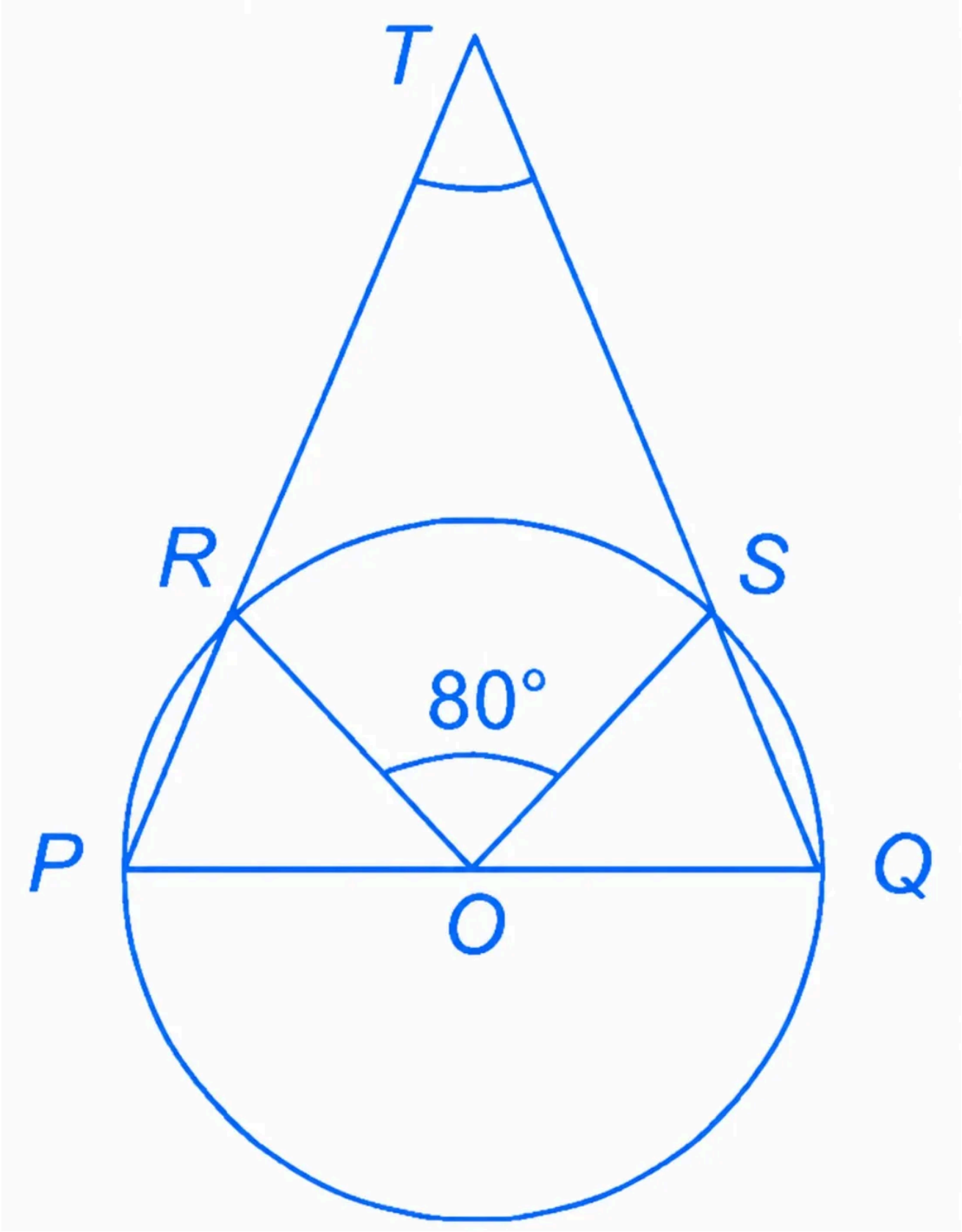
(2 Marks Questions)

Year: GATE EE 2026

Question 1: In the given figure, $\bar{PQ}$ is the diameter of a circle with center 0 . Two points R and S are chosen on the circle such that $\angle ROS = 80^{\circ}$. When $\bar{PR}$ and $\bar{QS}$ are extended, they meet at T . The value of $\angle RTS$ is _____ .

Are you liking this doc?

Yes No



A: 40°

B: 50°

C: 60°

D: 80°

Are you liking this doc?

Yes

No

Answer: B

Solution:

$\frac{1}{2}(180^\circ - 80^\circ) = 50^\circ$

Year: GATE EE 2026

Question 2: P and Q are two positive integers such that $P^2 = Q^2 + 13$. The product of the numbers P and Q is _____

A: 13

B: 26

C: 39

D: 42

Answer: D

Solution:

$$\begin{aligned} [P^2 - Q^2] &= 13 \\ (P-Q)(P+Q) &= 13 = 1 \times 13 \\ P - Q &= 1 \\ P + Q &= 13 \\ P &= 7 \\ and\ Q &= 6 \\ P \times Q &= 42 \end{aligned}$$

Year: GATE EE 2025

Question 3: Scores obtained by two students P and Q in seven courses are given in the table below. Based on the information given in the table, which one of the following statements is INCORRECT?

P	22	89	50	45	78	60	39
Q	35	65	60	56	81	45	50

Are you liking this doc?

Yes

No

A: Average score of P is less than the average score of Q.

B: Median score of P is same as the median score of Q.

C: Difference between the maximum and minimum scores of P is greater than the

difference between the maximum and minimum scores of Q.

D: Median score and the average score of Q are same.

Answer: B

Solution:

Median of $P = 50$

Median of $Q = 56$

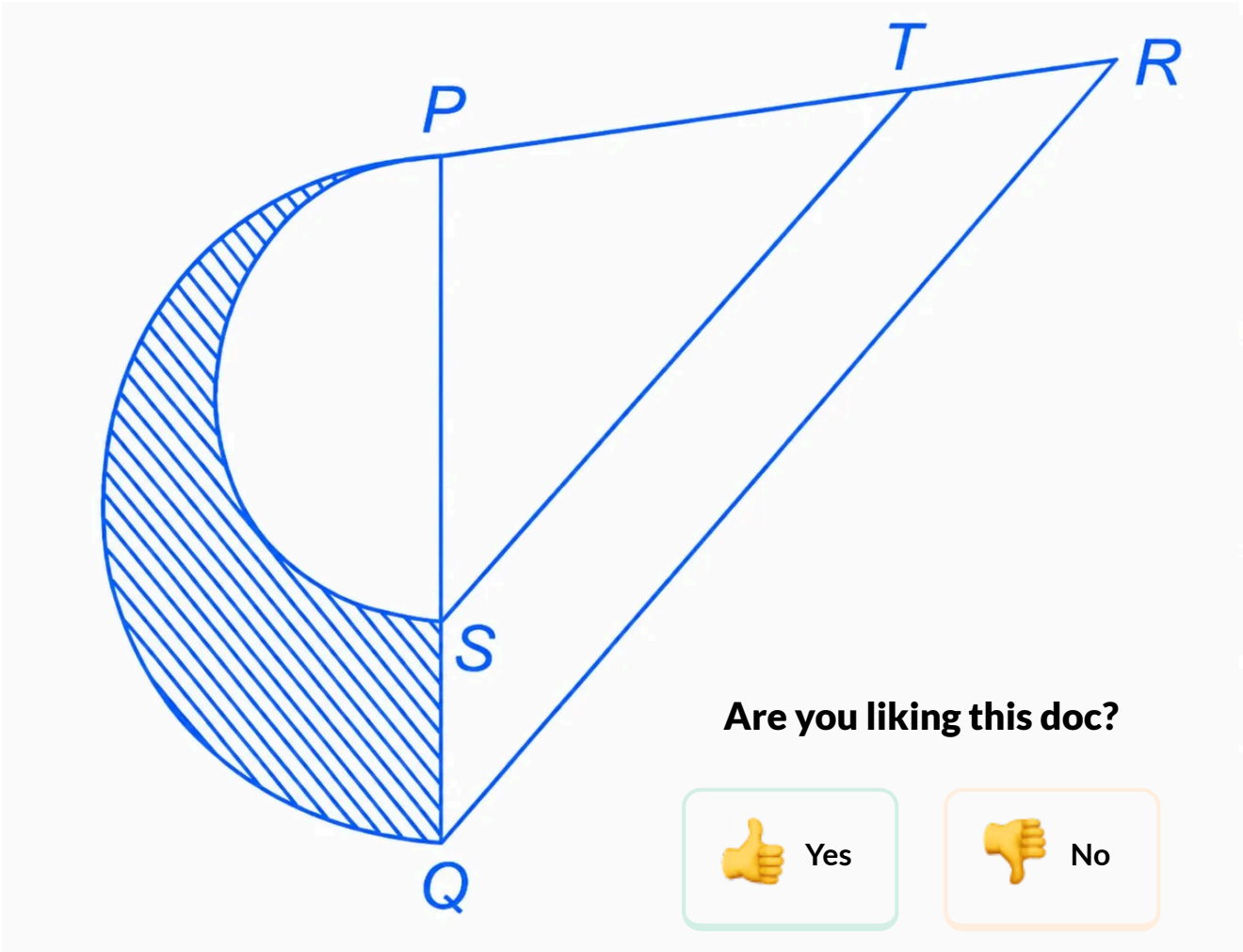
$\therefore$ Median of score of P is same as the median score of Q is incorrect.

Year: GATE EE 2025

Question 4: In the triangle PQR , the lengths of PT and TR are in the ratio of $3:2$. ST is parallel to QR. Two semi-circles are drawn with PS and PQ as diameters, as shown in the figure.

Which one of the following statements is true about the shaded area PQS?

Note: The figure shown is representative.



A: The shaded area is $\frac{16}{9}$ times the area of the semi-circle with the diameter PS.

B: The shaded area is equal to the area of the semi-circle with the diameter PS.

C: The shaded area is $\frac{14}{9}$ times the area of the semi-circle with the diameter PS.

D: The shaded area is $\frac{14}{25}$ times the area of the semi-circle with the diameter PQ.

Are you liking this doc?

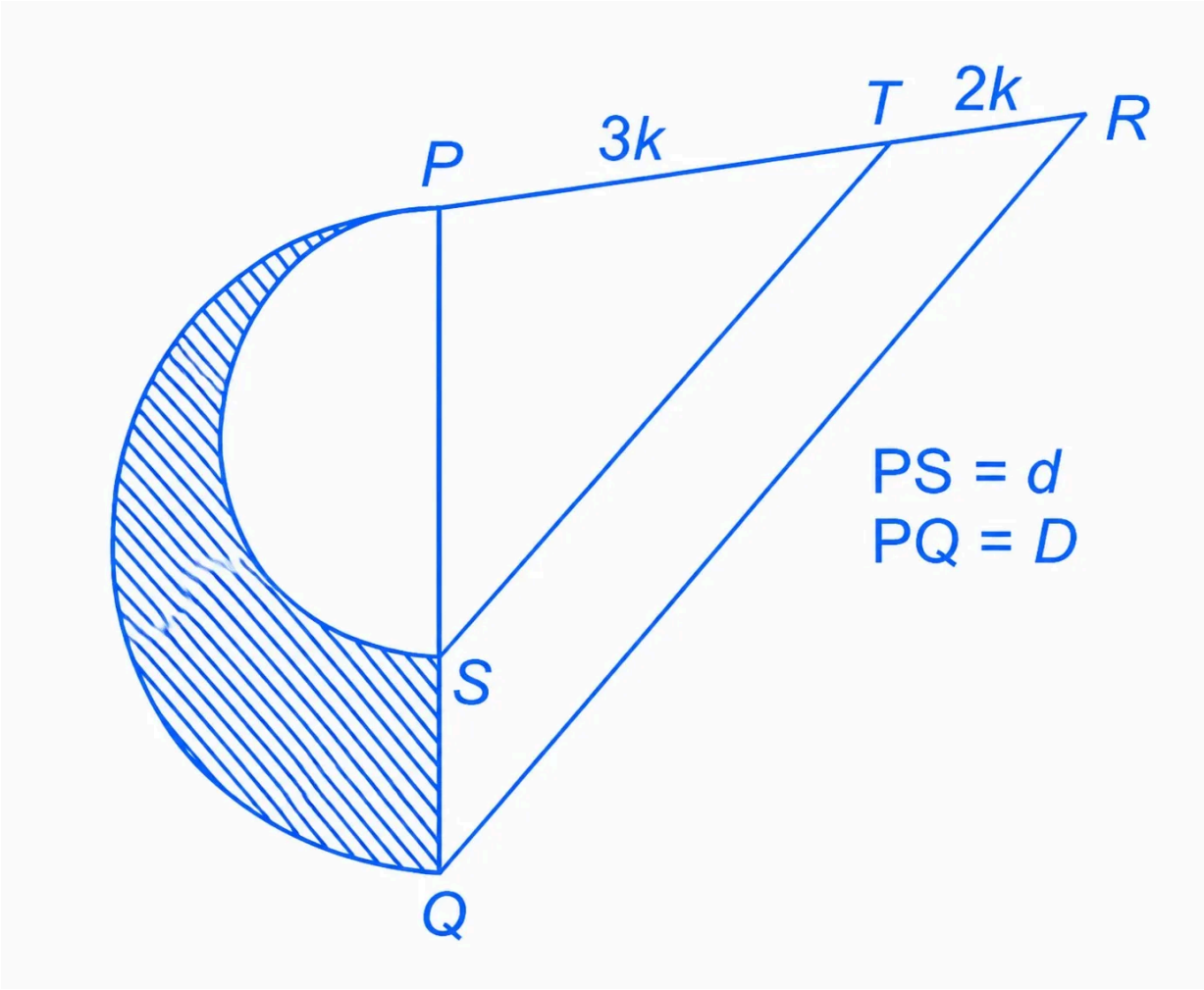
Yes

No

Answer: A

Solution:

$\mathrm{PT}: TR = 3:2$



$$\begin{aligned}
\Delta PST &\sim \Delta PQR \\
\frac{PT}{PR} &= \frac{PS}{PQ} \\
\frac{3k}{5k} &= \frac{PS}{PQ} \\
\frac{PS}{PQ} &= \frac{3k}{5k} = \frac{3}{5} \\
PQ &= \frac{5}{3}PS
\end{aligned}$$

$$\begin{aligned}
\text{Area of shaded} &= \text{Area of big semi-circle - Area of small semi-circle} \\
&= \frac{1}{2}\pi\left(\frac{D}{2}\right)^2 - \frac{1}{2}\pi\left(\frac{d}{2}\right)^2 \\
&= \frac{\pi}{8}(D^2 - d^2) = \frac{16}{9}\frac{\pi}{8}PS^2
\end{aligned}$$

Are you liking this doc?

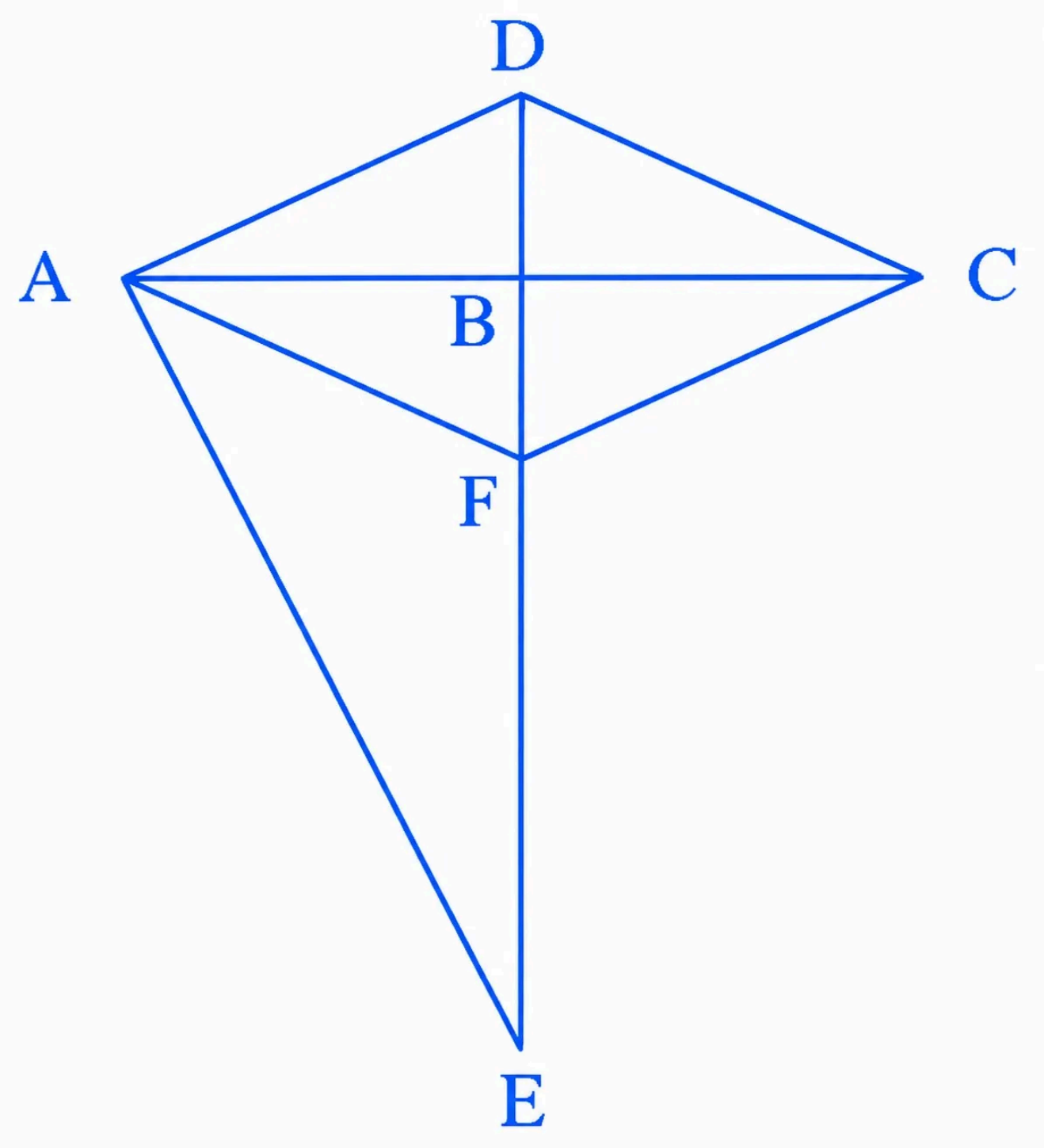
Yes No

Year: GATE EE 2024

Question 5: In the following figure,

- **CD = 5 cm, BE = 10 cm, AE = 12 cm,**
- **∠DAB = ∠DCB, and ∠DAE = ∠DBC = 90°**

Points AFCD create a rhombus.



The length of BF (in cm) is

A: 3

B: 2

C: 4

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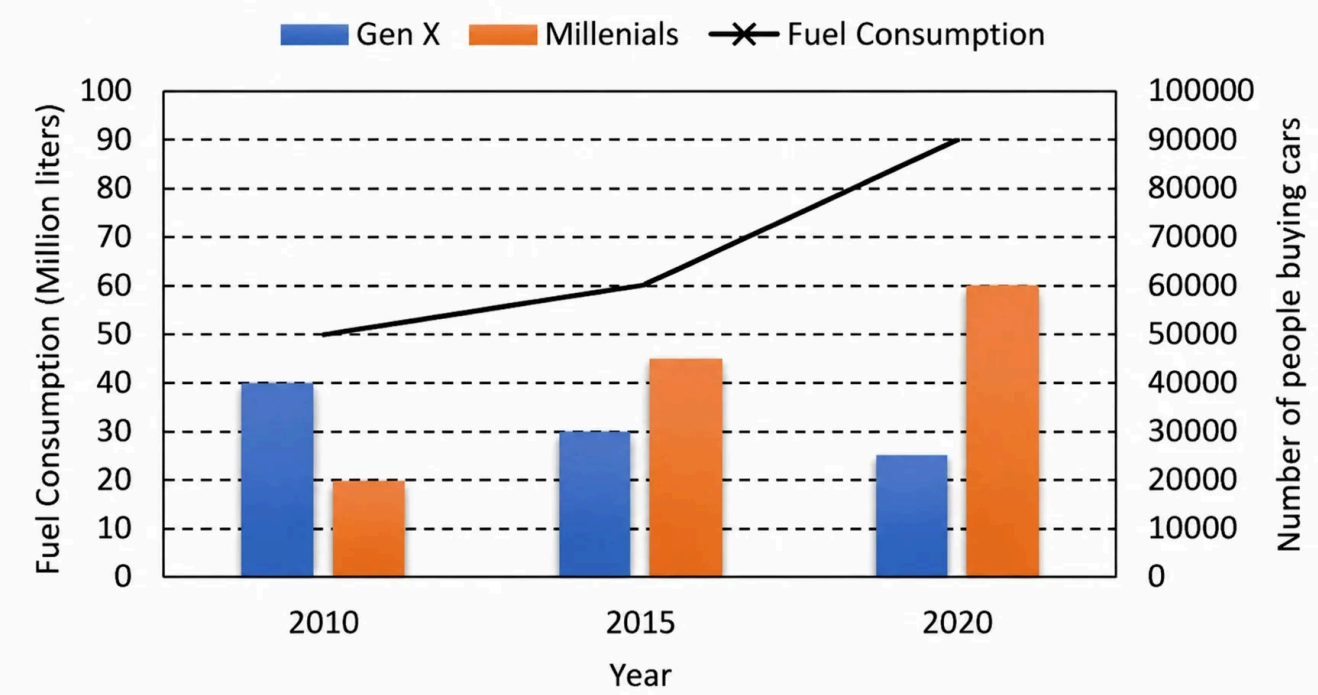
No

D: 6

Answer: A

Year: GATE EE 2024

Question 6: The chart below shows the data of the number of cars bought by Millennials and Gen X people in a country from the year 2010 to 2020 as well as the yearly fuel consumption of the country (in Million liters).



Considering the data presented in the chart, which one of the following options is true?

A: The percentage increase in fuel consumption from 2010 to 2015 is more than the percentage increase in fuel consumption from 2015 to 2020.

B: The increase in the number of Millennial car buyers from 2015 to 2020 is less than the decrease in the number of Gen X car buyers from 2010 to 2015.

C: The increase in the number of Millennial car buyers from 2010 to 2015 is more than the decrease in the number of Gen X car buyers from 2010 to 2015.

D: The decrease in the number of Gen X car buyers from 2015 to 2020 is more than the increase in the number of Millennial car buyers from 2010 to 2015.

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Yes No

Answer: C

Year: GATE EE 2024

Question 7: A surveyor has to measure the horizontal distance from her position to a distant reference point C. Using her position as the center, a 200 m horizontal line segment is drawn with the two endpoints A and B. Points A, B, and C are not collinear. Each of the angles ∠CAB and ∠CBA are measured as 87.8°. The distance (in m) of the reference point C from her position is nearest to

A: 2603

B: 2606

C: 2306

D: 2063

Answer: A

Year: GATE EE 2022

Question 8: There are two identical dice with a single letter on each of the faces. The following six letters : Q, R, S, T, U and V, one on each of the faces. Any of the six outcomes are equally likely.

The two dice are thrown once independently at random.

What is the probability that the outcomes on the dice were composed only of any combination of the following possible outcomes : Q, U and V?

A: $\frac{1}{4}$

B: $\frac{3}{4}$

C: $\frac{1}{6}$

D: $\frac{5}{36}$

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Yes No

Answer: A

Solution:

To calculate the probability that the outcomes on the dice are composed only of any combination of the letters Q, U, and V, we need to follow these steps:

1. Determine the total number of possible outcomes when the two dice are thrown.
2. Determine the number of favorable outcomes where the results involve only the letters Q, U, and V.
3. Calculate the probability by dividing the number of favorable outcomes by the total number of possible outcomes.

Step 1: Determine the total number of possible outcomes

Since each die has 6 faces, and there are two dice, the total number of possible outcomes is given by:

$6 \times 6 = 36$

Step 2: Determine the number of favorable outcomes

The favorable outcomes consist only of the letters Q, U, and V. Each die can display any of these 3 letters. Therefore, the number of favorable outcomes is given by:

$3 \times 3 = 9$

Step 3: Calculate the probability

The probability of getting a favorable outcome is the ratio of the number of favorable outcomes to the total number of possible outcomes:

$$\text{Probability} = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}} = \frac{9}{36}$$

This fraction can be simplified as follows:

$$\frac{9}{36} = \frac{1}{4}$$

Therefore, the probability that the outcomes on the dice were composed only of any combination of the letters Q, U, and V is $\frac{1}{4}$.

Option A: $\frac{1}{4}$

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Yes No

Year: GATE EE 2022

Question 9: The price of an item is 10% cheaper in an online store S compared to the price at another online store M. Store S charges Rs.150 for delivery. There are no delivery charges for orders from the store M. A person bought the item from the store S and saved Rs.100. What is the price of the item at the online store S (in Rs.) if there are no other charges than what is described above?

A: 2500

B: 2250

C: 1750

D: 1500

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Yes

No

Answer: B

Solution:

Let's denote the price of the item at store M as P (in Rs.). Since the item at store S is 10% cheaper than at store M, the price at store S is:

$P_S = P - 0.10P = 0.90P$

Store S also charges Rs.150 for delivery. Hence, the total cost when buying from store S is:

$\text{Total cost at store S} = 0.90P + 150$

According to the problem, the person saved Rs.100 by buying from store S. This means that the total cost at store M is Rs.100 more than the total cost at store S:

$P = 0.90P + 150 + 100$

Simplifying this equation:

$P = 0.90P + 250$

Subtracting $0.90P$ from both sides of the equation:

$0.10P = 250$

Solving for P:

$P = \frac{250}{0.10} = 2500$

Thus, the price of the item at store M is Rs.2500. The price of the item at store S (without including the delivery charge) is:

$P_S = 0.90 \times 2500 = 2250$

Therefore, the price of the item at the online store S is Rs.2250.

Answer: Option B - 2250

Year: GATE EE 2022

Question 10: The letters P, Q, R, S, T and U are to be placed one per vertex on a regular convex hexagon, but not necessarily in the same order. Consider the following statements :

- **The line segment joint R and S is longer than the line segment joining P and Q.**
- **The line segment joining R and S is perpendicular to the line segment joining P and Q.**

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- **The line segment joining R and U is parallel to the line segment joining T and Q.**

Based on the above statements, which one of the following options is CORRECT?

A: The line segment joining R and T is parallel to the line segment joining Q and S.
B: The line segment joining T and Q is parallel to the line joining P and U.
C: The line segment joining R and P is perpendicular to the line segment joining U and Q.
D: The line segment joining Q and S is perpendicular to the line segment joining R and P.

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Yes

No

Answer: A

Solution:

Given :

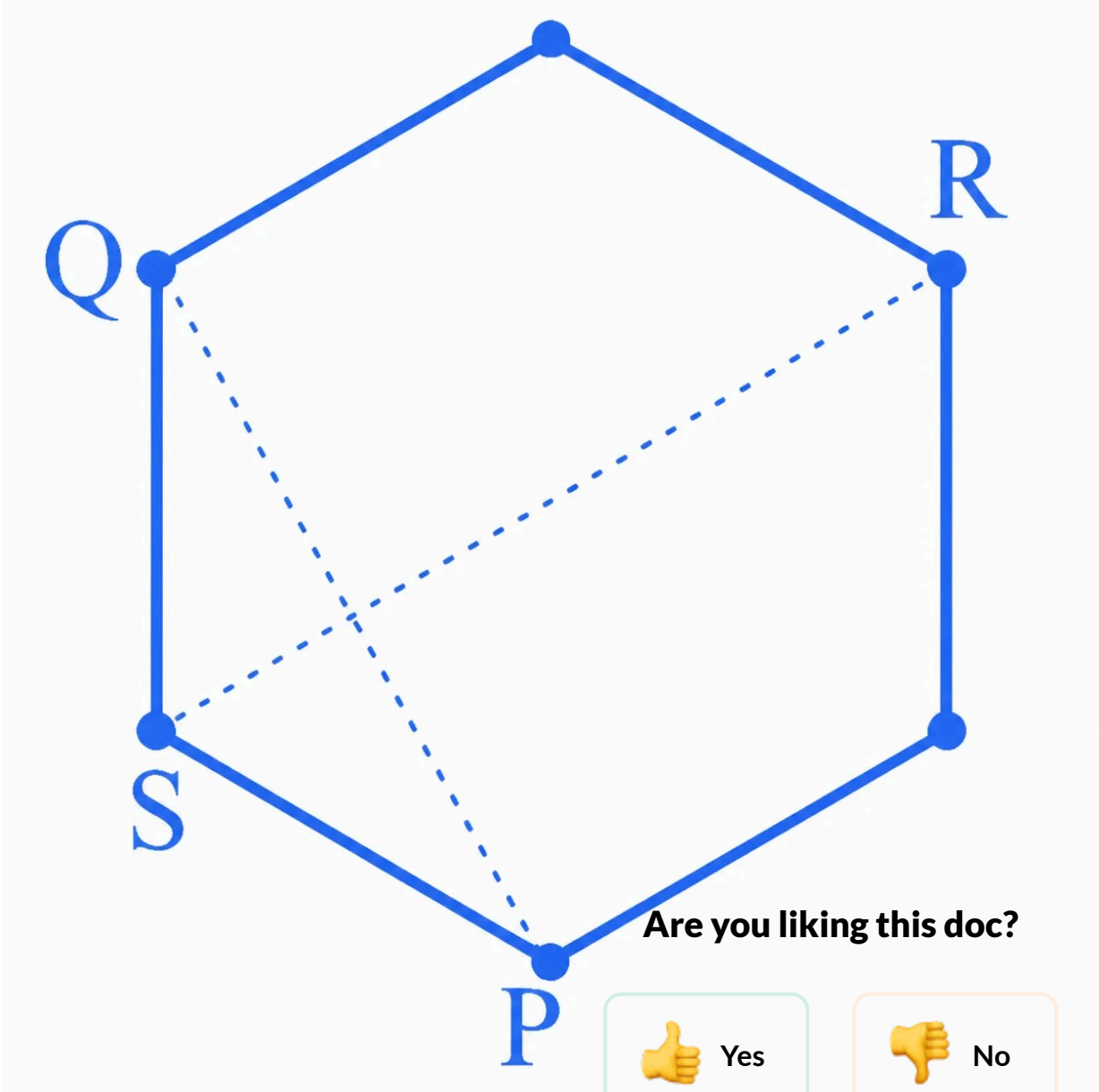
The letters P, Q, R, S, T, and U are placed one per vertex on a regular convex hexagon. The given conditions are as follows:

The line segment joining R and S is longer than the line segment joining P and Q.

$RS > PQ$

The line segment joining R and S is perpendicular to the line segment joining P and Q.

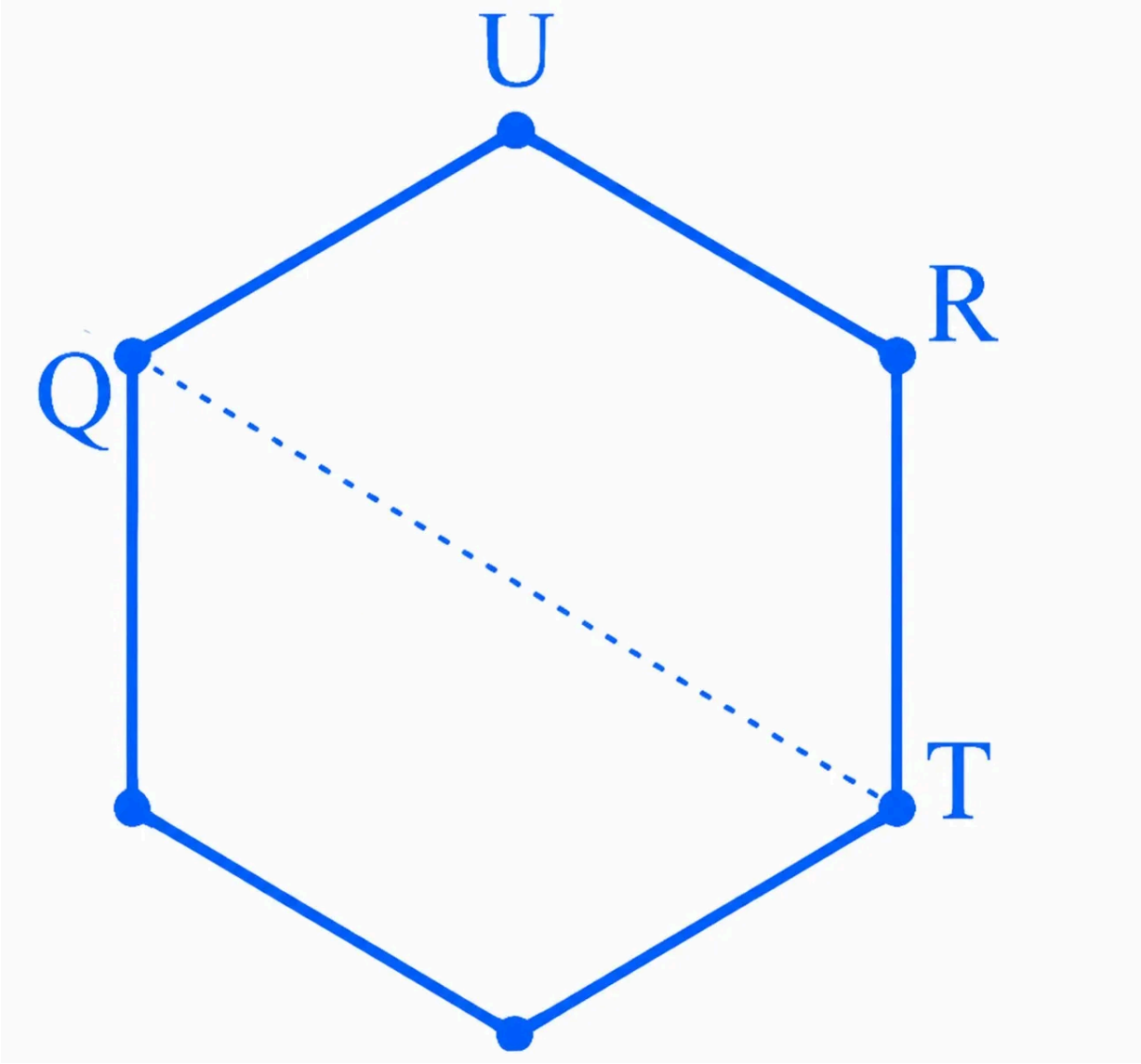
$RS \perp PQ$



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No

The line segment joining R and U is parallel to the line segment joining T and Q.

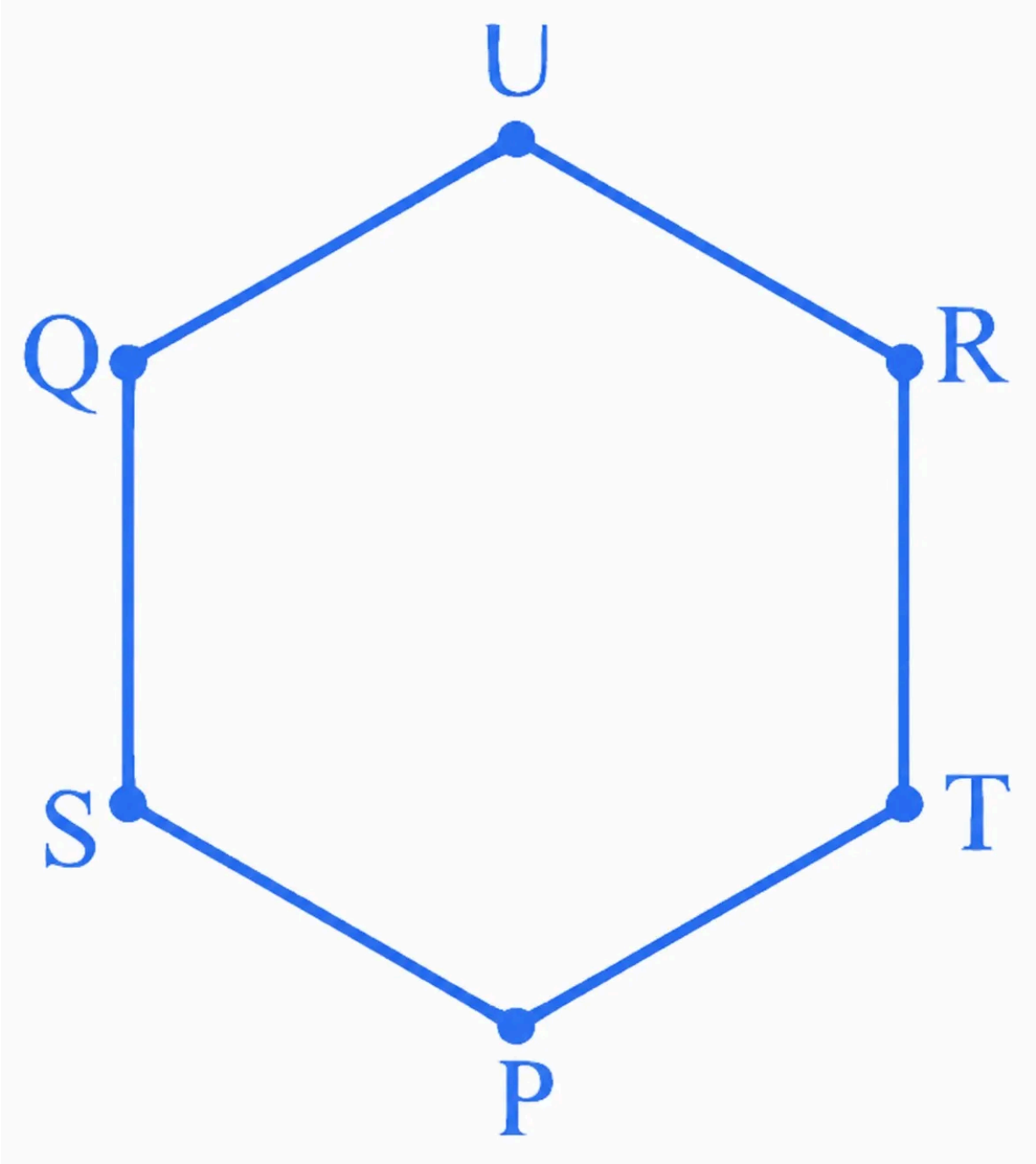
$RU \| TQ$



According to the given conditions, we can draw a diagram as shown below:

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Yes No

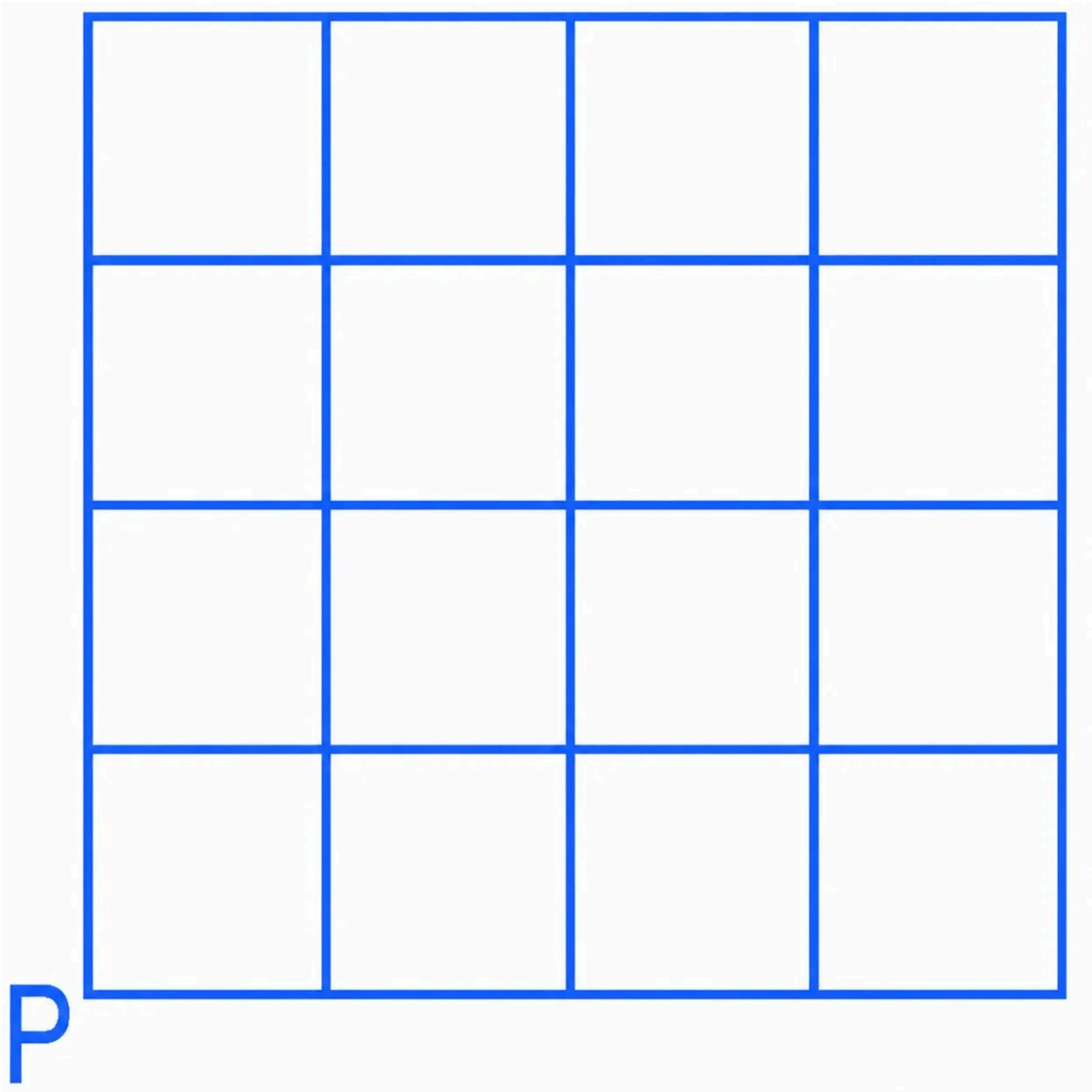


Based on the diagram and the given conditions, Option (A) is correct as the line segment RT is parallel to the line segment QS.

Year: GATE EE 2022

Question 11: An ant is at the bottom-left corner of a grid (point P) as shown below in the figure. It aims to move to the top-right corner of the grid. The ant moves only along the lines marked in the grid such that the current distance to the top-right corner strictly decreases. Which of the following is/are part of a possible trajectory of the ant during the movement?

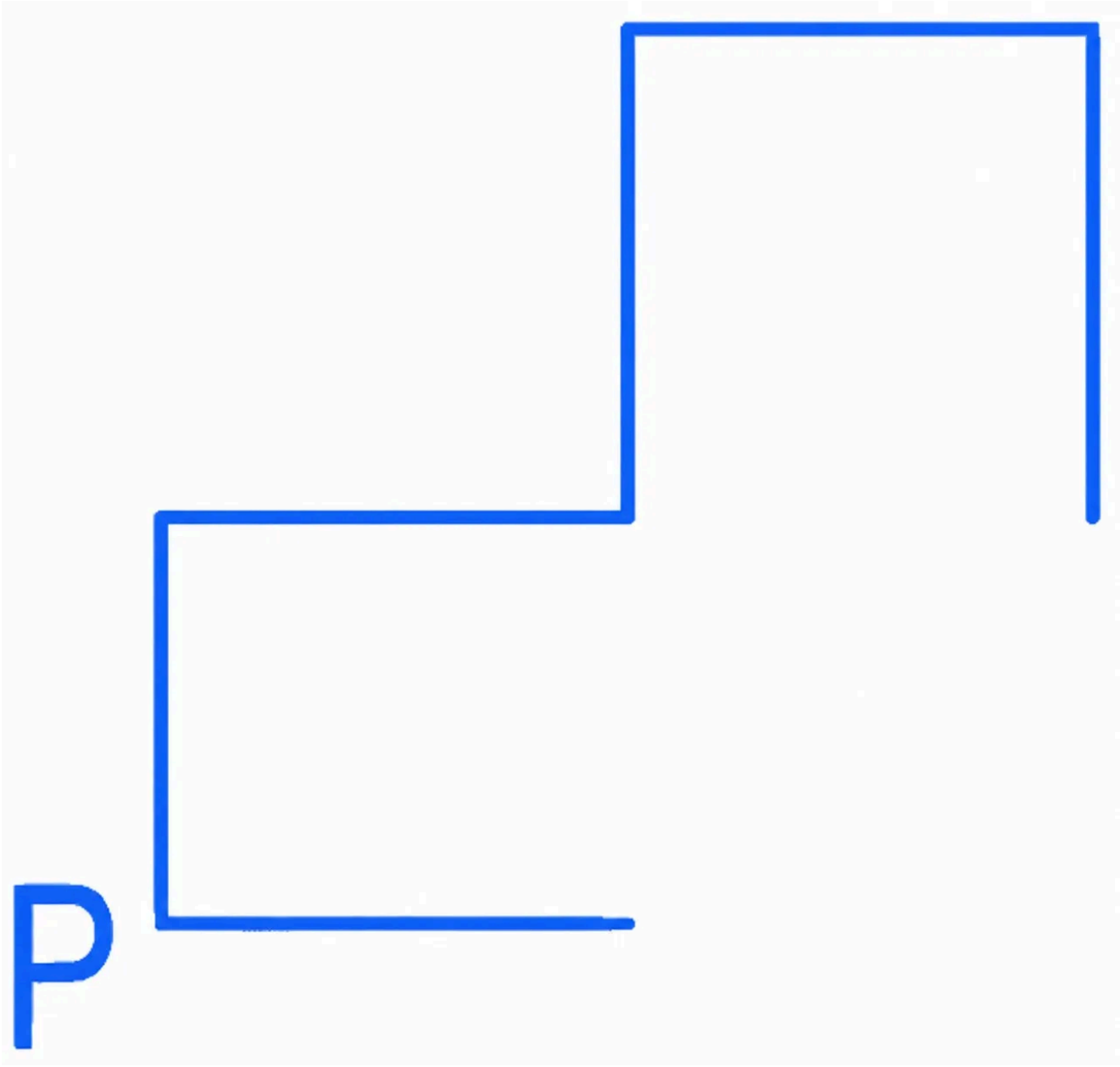
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Yes No

A:

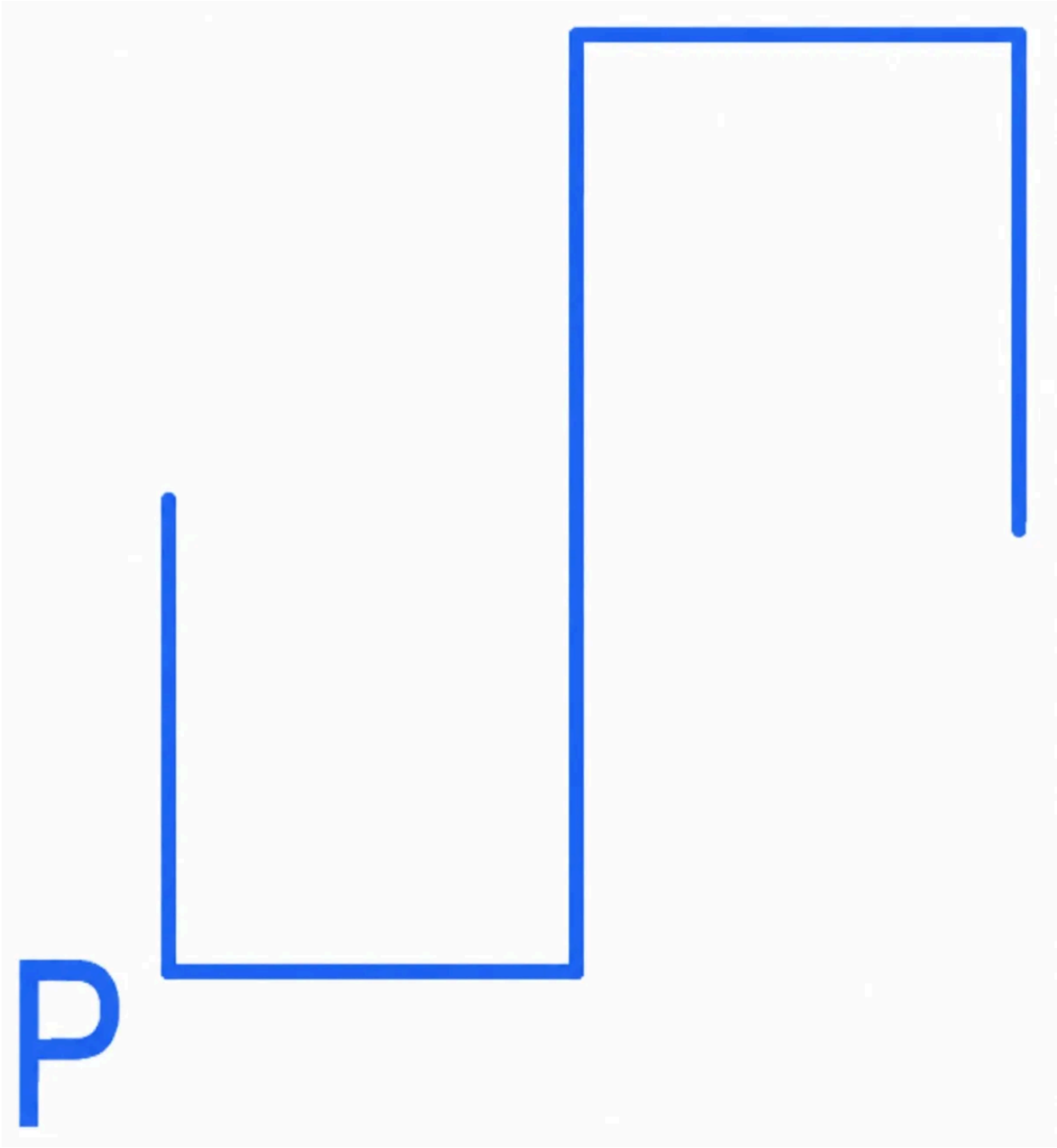


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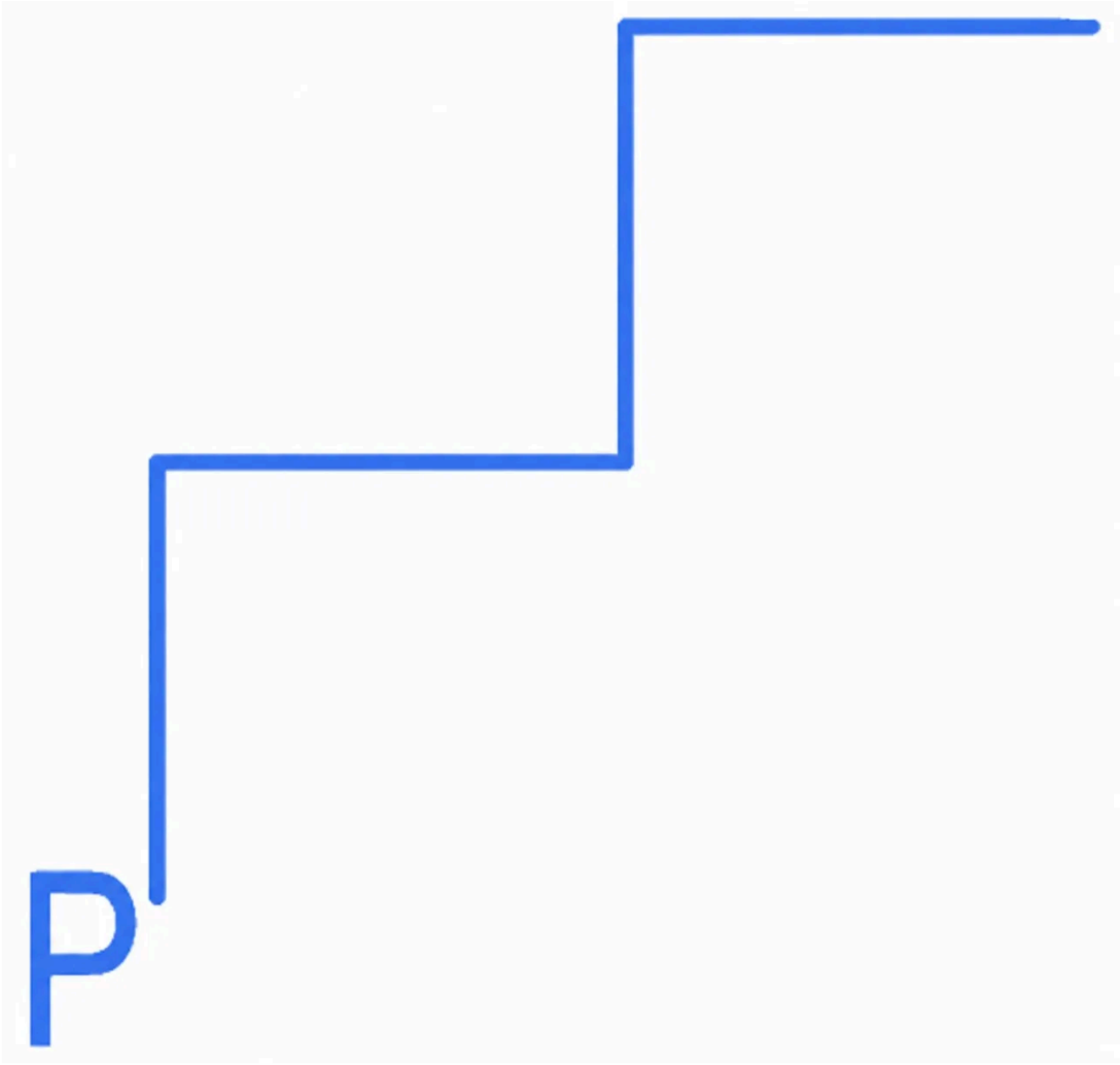
B:



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Yes No

C:

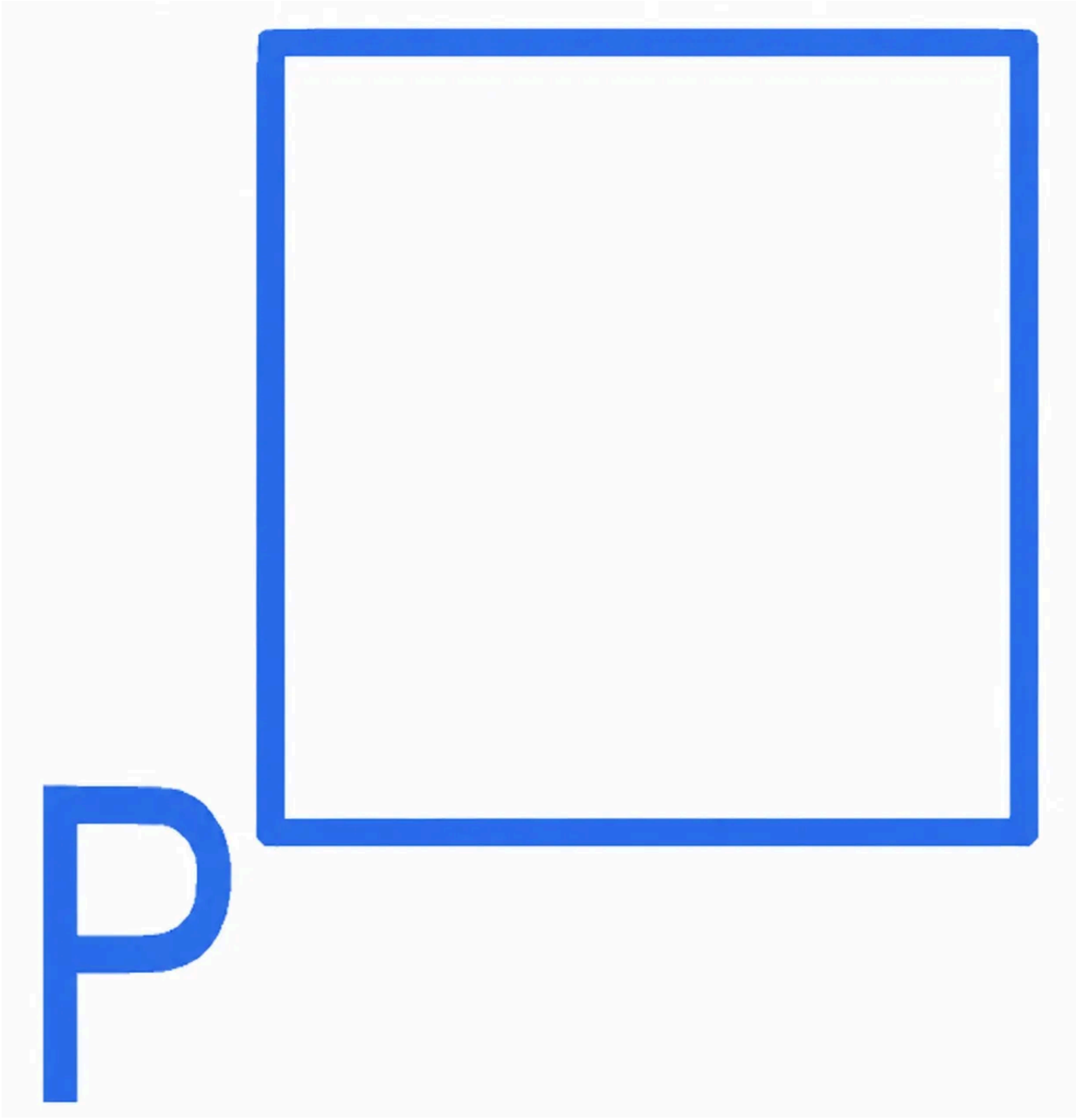


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Yes

No

D:



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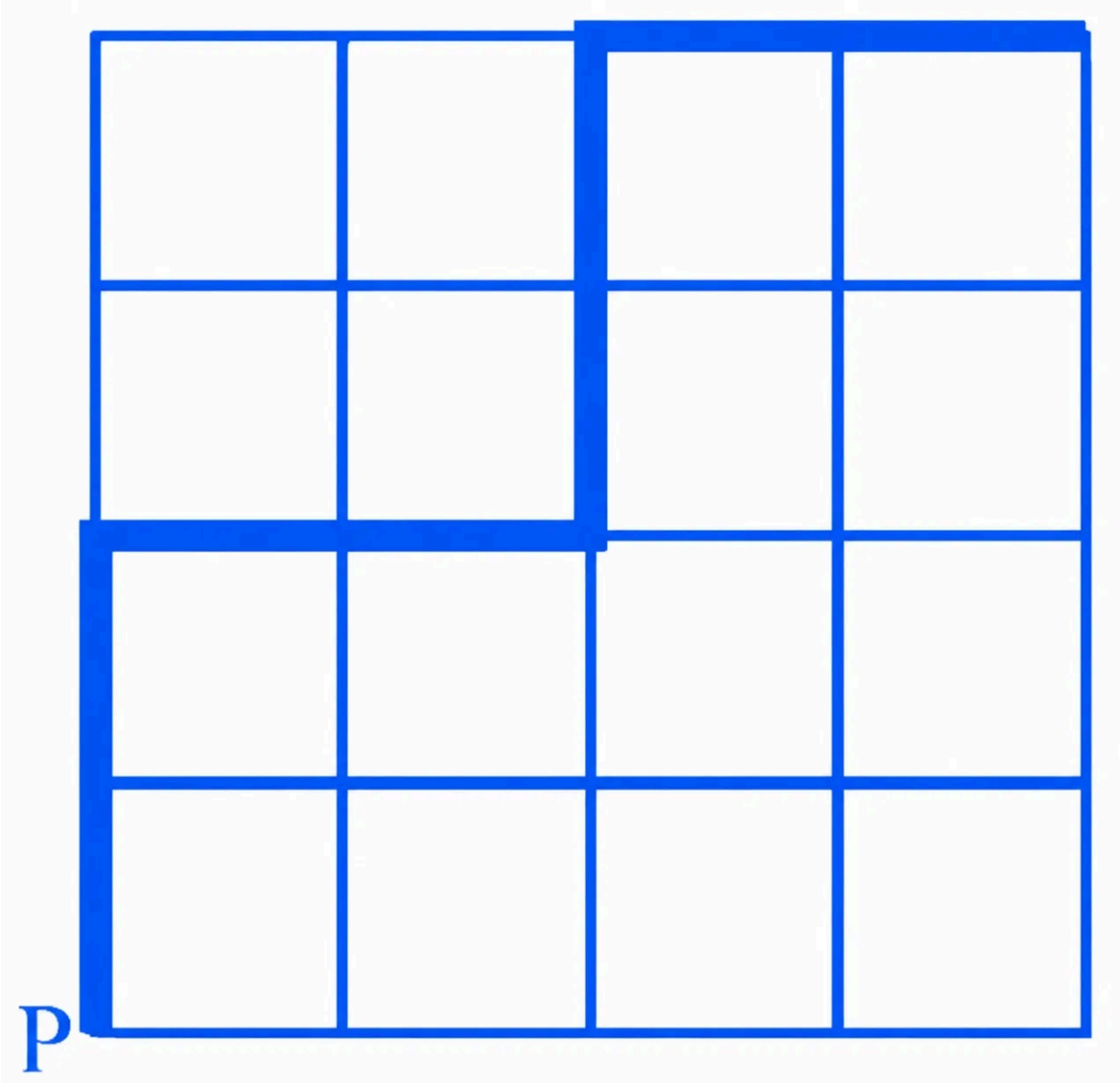
Yes

No

Answer: C

Solution:

The ant is positioned at the bottom-left corner of the grid, labeled as point P, as shown in the image below:



To reach the top-right corner, the ant must move along the grid lines, ensuring each move reduces the total distance to the top-right corner.

Upon analysis, the correct path that the ant can follow is illustrated in Option (C).

Therefore, the correct option is (C).

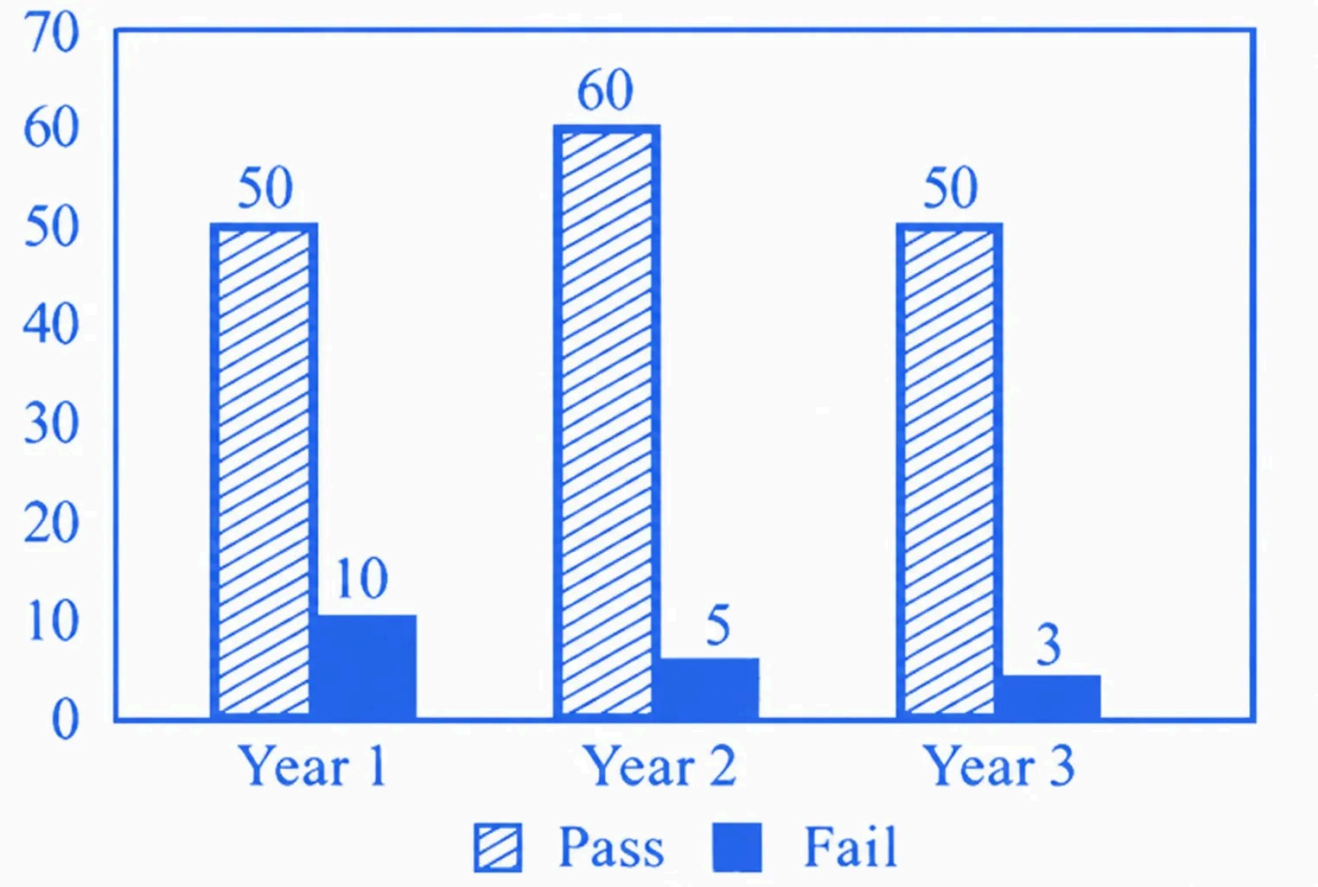
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Yes No

Year: GATE EE 2021

Question 12: The number of student passing or failing in an exam for a particular subject are presented in the bar chart below. Students who pass the exam cannot

appear for the exam again. Students who fails the exam in the first attempt must appear for the exam in the following year. Students always pass the exam in their second attempt. The numbers of students who took the exam for the first time in the year 2nd the year 3 respectively, are _____ .



A: 65 and 53

B: 60 and 50

C: 55 and 53

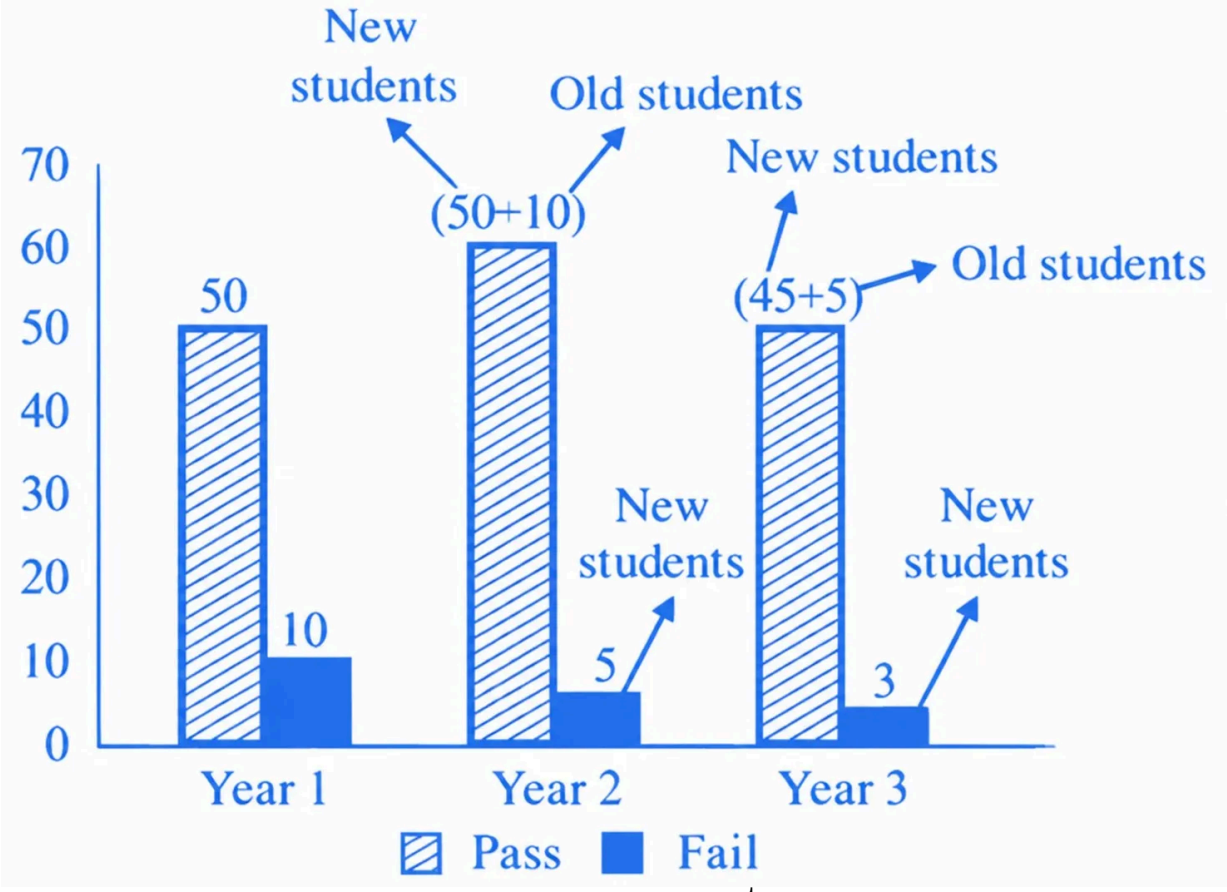
D: 55 and 48

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Yes No

Answer: D

Solution:



$\therefore$ The number of students who took exam for 1st time in year $2 \Rightarrow 50 + 5 = 55$

The number of students who took exam for 1st time in year $3 \Rightarrow 45 + 3 = 48$

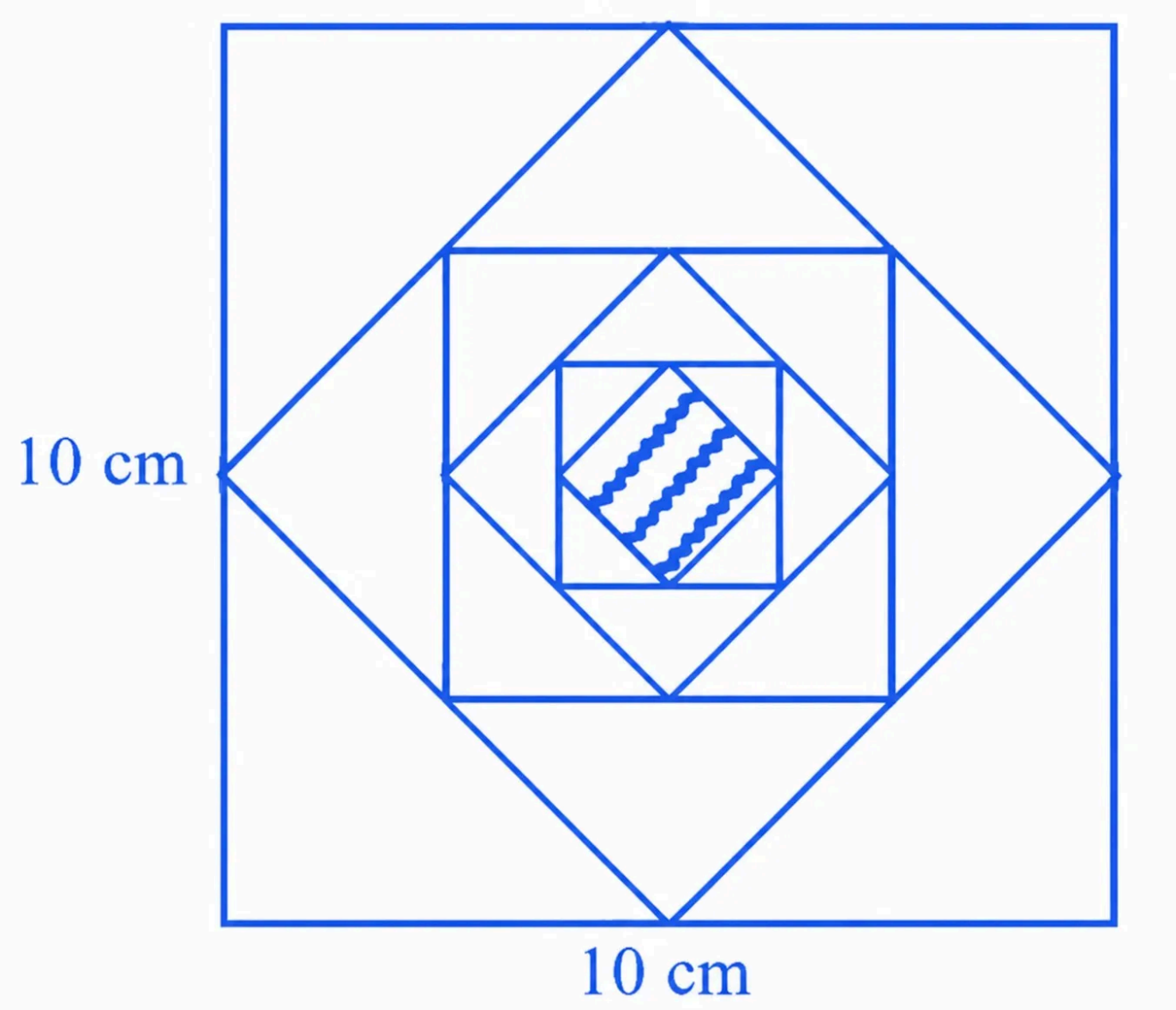
Hence, the correct option is (D).

Year: GATE EE 2021

Question 13: In the figure shown above, each inside square is formed by joining the mid points of the sides of the next larger square. The area of the smallest square (shaded) as shown, in cm^2 is

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A: 12.50

B: 6.25

C: 3.125

D: 1.5625

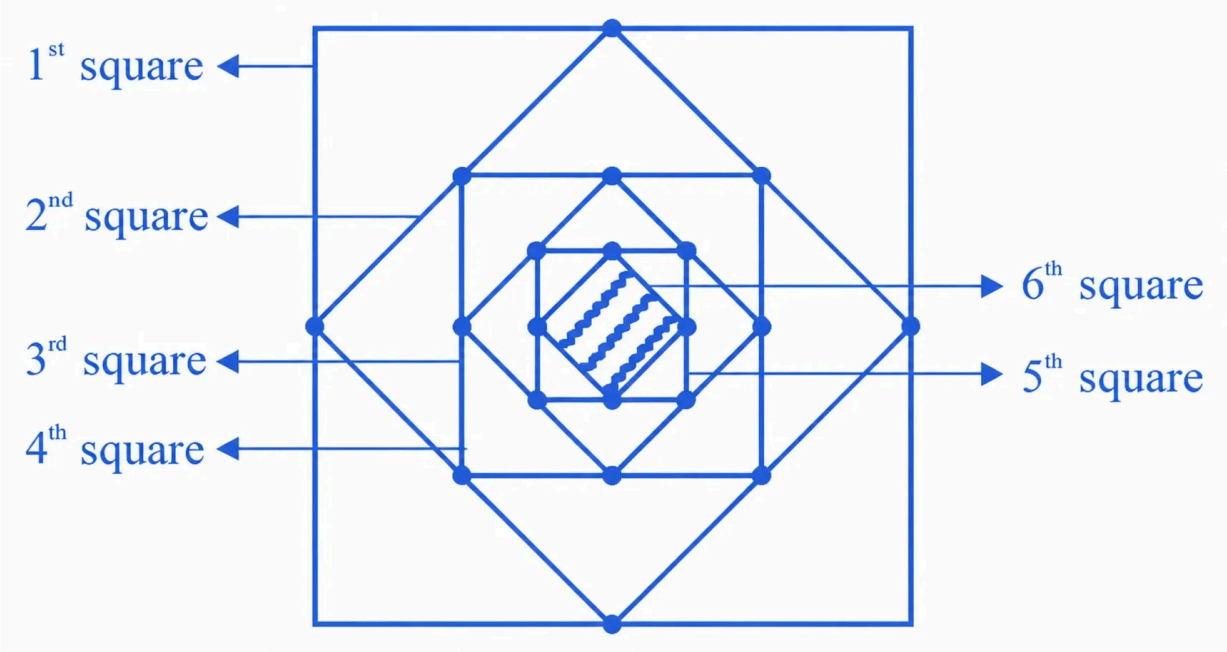
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Yes

No

Answer: C

Solution:



From figure, side of outer square $=$ diagonal of inside square

$\therefore$ Side of square $\Rightarrow$

$$\because \text{Side of } 6^{\text{th}} \text{ square (smallest square)} = \frac{10}{4\sqrt{2}}$$

$$\therefore \text{Area } = \frac{100}{32} = 3.125\ cm^2$$

Hence, the correct option is (C).

Year: GATE EE 2021

Question 14: Let X is a continuous random variable denoting the temperature measured. The range of temperature is $[0, 100]$ degree Celsius and let the probability density function of X be $f(x) = 0.01$ for $0 \leq X \leq 100$. The mean of X is ____ .

A: 2.5

B: 5.0

C: 25.0

D: 50.0

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Yes No

Answer: D

Solution:

Given : PDF $f(x) = 0.01; 0 \leq x \leq 100$

$E(x) = \int_{-\infty}^{\infty} xf(x)dx$

$\because$ It is uniform probability distribution

$\therefore E(x) = \frac{0+100}{2} = 50$

Hence, the correct option is (D).

Year: GATE EE 2019

Question 15: How many integers are there between 100 and 1000 all of whose digits are even?

A: 100

B: 90

C: 60

D: 80

Answer: A

Year: GATE EE 2019

Question 16: The ratio of the number of boys and girls who participated in an examination is 4 : 3. The total percentage of candidates who passed the examination is 80 and the percentage of a girls who passed is 90. The percentage of boys who passed is _____.

A: 90.00

B: 80.50

C: 55.50

D: 72.50

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Answer: D

Yes No

Year: GATE EE 2019

Question 17: Given two sets X = {1, 2, 3} and Y = {2, 3, 4}, we construct a set Z of all possible fractions where the numerators belong to set X and the denominators belong to set Y. The product of elements having minimum and maximum values in the set Z is _____.

A: $\frac{1}{12}$

B: $\frac{1}{8}$

C: $\frac{1}{6}$

D: $\frac{3}{8}$

Answer: D

Year: GATE EE 2018

Question 18: An e-mail password must contain three characters. The password has to contain one numeral from 0 to 9, one upper case and one lower case character from the English alphabet. How many distinct passwords are possible?

A: 6,760

B: 13,520

C: 40,560

D: 1,05,456

Answer: C

Year: GATE EE 2018

Question 19: A class of twelve children has two more boys than girls. A group of three children are randomly picked from this class to accompany the teacher on a field trip. What is the probability that the group accompanying the teacher contains more girls than boys?

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No

A: 0

B: $\frac{325}{864}$

C: $\frac{525}{864}$

D: $\frac{5}{12}$

Answer: B

Year: GATE EE 2018

Question 20: A designer uses marbles of four different colours for his designs. The cost of each marble is the same, irrespective of the colour. The table below shows the percentage of marbles of each colour used in the current design. The cost of each marble increased by 25%. Therefore, the designer decided to reduce equal numbers of marbles of each colour to keep the total cost unchanged. What is the percentage of blue marbles in the new design?

Blue	Black	Red	Yellow
40%	25%	20%	15%

A: 35.75

B: 40.25

C: 43.75

D: 46.25

Answer: C

Year: GATE EE 2018

Question 21: P, Q, R and S crossed a lake in a boat that can hold a maximum of two persons, with only one set of oars. The following additional facts are available.

(i) The boat held two persons on each of the three forward trips across the lake and one person on each of the two return trips.

(ii) P is unable to row when someone else is in the boat.

(iii) Q is unable to row with anyone else except R.

(iv) Each person rowed for at least one trip.

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(v) Only one person can row during a trip.

Who rowed twice?

A: P

B: Q

C: R

D: S

Answer: C

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